

SOD1 is a synthetic lethal target in *PPM1D*-mutant leukemia cells

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2

April 4, 2024 (this version)

Reviewed preprint version 1

October 4, 2023

Posted to preprint server

September 1, 2023

Sent for peer review

August 11, 2023

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Abstract

The DNA damage response is critical for maintaining genome integrity and is commonly disrupted in the development of cancer. *PPM1D* (protein phosphatase, Mg²⁺/Mn²⁺ dependent 1D) is a master negative regulator of the response; gain-of-function mutations and amplifications of *PPM1D* are found across several human cancers making it a relevant pharmacologic target. Here, we used CRISPR/Cas9 screening to identify synthetic-lethal dependencies of *PPM1D*, uncovering superoxide dismutase-1 (SOD1) as a potential target for *PPM1D*-mutant cells. We revealed a dysregulated redox landscape characterized by elevated levels of reactive oxygen species and a compromised response to oxidative stress in *PPM1D*-mutant cells. Altogether, our results demonstrate the protective role of SOD1 against oxidative stress in *PPM1D*-mutant leukemia cells and highlight a new potential therapeutic strategy against *PPM1D*-mutant cancers.

eLife assessment

Gain-of-function mutations and amplifications of PPM1D are found across several human cancers and are associated with advanced tumor stage and worse prognosis. Thus far, the clinical translation has not been possible due to the lack of PPM1D inhibitors with favorable pharmacokinetic properties. This **useful** study leverages CRISPR/Cas9 screening to determine that loss of SOD1 and is synthetic lethal with PPM1D mutation in leukemia. The mechanistic analyses at still **incomplete**.

Introduction

Cellular DNA is frequently damaged by both endogenous and exogenous factors (Hoeijmakers, 2009). Unresolved DNA damage can lead to genomic instability, which is a hallmark of aging and cancer (Hanahan and Weinberg, 2011). Cells have evolved intricate mechanisms to detect and repair DNA lesions. The DNA damage response (DDR) is a complex network of signaling pathways that coordinate various cellular processes initiated by p53, such as DNA repair (Ciccia and Elledge, 2010), cell cycle checkpoint activation (Harper et al., 1993), and apoptosis (Yonish-Rouach et al., 1991). However, upon resolution of DNA damage, the cell must terminate the DDR to avoid prolonged cell cycle arrest and apoptosis. One critical mechanism for DDR termination is the expression of Protein Phosphatase Mg^{2+}/Mn^{2+} -Dependent 1D (PPM1D) (Fiscella et al., 1997), which is induced by p53 and plays a key role in attenuating the response. PPM1D is a member of the PP2C family of serine/threonine protein phosphatases and has been shown to dephosphorylate a wide range of DDR signaling molecules including p53, p38 MAPK, CHK1, CHK2, and H2AX (Bulavin et al., 2002; Cha et al., 2010; Lu et al., 2005; Oliva-Trastoy et al., 2007; Takekawa et al., 2000). These dephosphorylation events generally lead to reduced activity of the targets, ultimately resulting in deactivation of the DDR.

Dysregulation of PPM1D has been associated with the development of diverse cancers, including breast, ovarian, esophagus, brain, and others (Khadka et al., 2022; Li et al., 2002; Li et al., 2020b; Ruark et al., 2013; Zhang et al., 2014). *PPM1D* is located on chromosome 17q and therefore frequently amplified in breast and ovarian cancers exhibiting 17q23 amplifications (Li et al., 2002; Ruark et al., 2013). These amplifications result in overexpression of the wildtype PPM1D protein and consequently leads to suppression of p53 and other PPM1D targets in the DDR (Bulavin et al., 2002; Lambros et al., 2010). In addition, PPM1D can also become dysregulated through mutations in its terminal exon. These mutations produce a truncated protein that is stabilized, evading proteasome-mediated degradation (Tokheim et al., 2021). The resulting mutant protein maintains its phosphatase activity and is found at high levels even in the absence of DNA damage. Excessive PPM1D activity leads to constitutive dephosphorylation and downregulation of PPM1D targets including multiple members of the DDR (Hsu et al., 2018). These gain-of-function *PPM1D* mutations are observed in diverse solid cancers including osteosarcoma (He et al., 2021), colorectal carcinoma (Peng et al., 2014; Yin et al., 2013), diffuse midline gliomas (Wang et al., 2011; Zhang et al., 2014) and others. Moreover, *PPM1D* mutations and overexpression are associated with advanced tumor stage, worse prognosis, and increased lymph node metastasis (Fu et al., 2014; Jiao et al., 2014; Li et al., 2020a; Li et al., 2020b; Peng et al., 2014; Zhang et al., 2014).

More recently, *PPM1D*-mutations have been shown to drive expansion of hematopoietic stem cells (Bolton et al., 2020; Hsu et al., 2018; Kahn et al., 2018) in association with clonal hematopoiesis (CH), a pre-malignant state associated with an increased risk of hematologic malignancies and elevated all-cause mortality (Genovese et al., 2014; Jaiswal et al., 2014).

PPM1D-mutations are particularly enriched in patients with prior exposure to cytotoxic therapies, who have a high risk of therapy-related myeloid neoplasms (t-MN) (Hsu et al., 2018 [↗](#); Lindsley et al., 2017 [↗](#)). Given the prevalence of *PPM1D* aberrations in cancer, *PPM1D* is an attractive therapeutic target. Ongoing efforts are focused on elucidating the structure of *PPM1D* to improve drug design and development (Miller et al., 2022 [↗](#)). While several inhibitors thus far have shown efficacy *in vitro*, few have been studied *in vivo* and none have progressed to clinical trials due to poor bioavailability. Therefore, identifying targetable, synthetic-lethal partners to exploit the genetic defects of *PPM1D*-altered cells can offer an alternative therapeutic approach.

In this study, we performed an unbiased, whole-genome CRISPR screen to investigate genes essential for cell survival in *PPM1D*-mutated leukemia cell lines. We identified superoxide dismutase-1 (*SOD1*) as a novel synthetic-lethal dependency of *PPM1D* which was validated by genetic and pharmacologic approaches. We showed that the mutant cells display compromised responses to oxidative stress and DNA damage, leading to increased reactive oxygen species and genomic instability. These results provide valuable insights into the biological processes corrupted by mutant *PPM1D* and underscore the potential of *SOD1* as a targetable vulnerability in this context.

Results

SOD1 is a synthetic lethal vulnerability of *PPM1D*-mutant leukemia cells

CRISPR dropout screens have emerged as a powerful tool to assess the functional importance of individual genes within a particular pathway by measuring the impact of their depletion on cell viability or fitness. To identify genes essential for *PPM1D*-mutant cell survival, we first created isogenic wild-type (WT) and *PPM1D*-mutant Cas9-expressing OCI-AML2 leukemia cell lines and selected two *PPM1D*-mutant clones for CRISPR screening (**Figure 1-figure supplement 1A** [↗](#)). We transduced the cells with a whole-genome lentiviral library containing 90,709 guide RNAs (gRNAs) targeting 18,010 human genes (Tzelepis et al., 2016 [↗](#)). At day ten post-transduction, the cells were harvested for the first timepoint and then subsequently passaged for an additional 18 days to allow for negatively selected gene-knockout cells to “drop out”. The remaining pool of cells were collected for deep sequencing analysis of gRNA abundance (**Figure 1A** [↗](#)). We analyzed genes that were specifically depleted in the mutant but not WT cells using the MaGECK-VISPR pipeline (Li et al., 2014 [↗](#)). Differentially depleted genes are those for which the knockout or depletion of the gene results in a significant impact on the viability or growth of *PPM1D*-mutant cells compared to WT control cells. Through this analysis, we identified 409 differentially depleted genes in one of the *PPM1D*-mutant clones and 92 differentially depleted genes in the other clone while adhering to the maximum false discovery rate (FDR) cutoff of 25%. Among these genes, we found 37 common candidates that were depleted in both *PPM1D*-mutant biological replicates that were not depleted in the WT control cells (**Figure 1-figure supplement 1B** [↗](#), **Figure 1-source data 1**).

Gene ontology analysis of these top essential genes demonstrated an enrichment in pathways related to DNA repair, interstrand crosslink (ICL) repair, and cellular responses to stress (**Figure 1B** [↗](#)). Pathway analyses with the KEGG and REAC databases revealed a significant enrichment of the Fanconi anemia (FA) repair pathway, with notable genes such as *BRIP1* (*FANCF*), *FANCI*, *FANCA*, *SLX4* (*FANCP*), *UBE2T* (*FANCT*), and *C19orf40* (*FAAP24*) (**Figure 1-figure supplement 1C**). Interestingly, our dropout screen revealed that superoxide dismutase [Cu/Zn], or *SOD1*, was the top essential gene based on fitness score (**Figure 1C** [↗](#)). *SOD1* is a crucial enzyme involved in scavenging superoxide (O_2^-) radicals, which are harmful byproducts of mitochondrial cellular metabolism. Excessive reactive oxygen species (ROS) causes oxidative stress, which can damage cellular structures including DNA, proteins, and lipids. *SOD1* is an attractive therapeutic target due

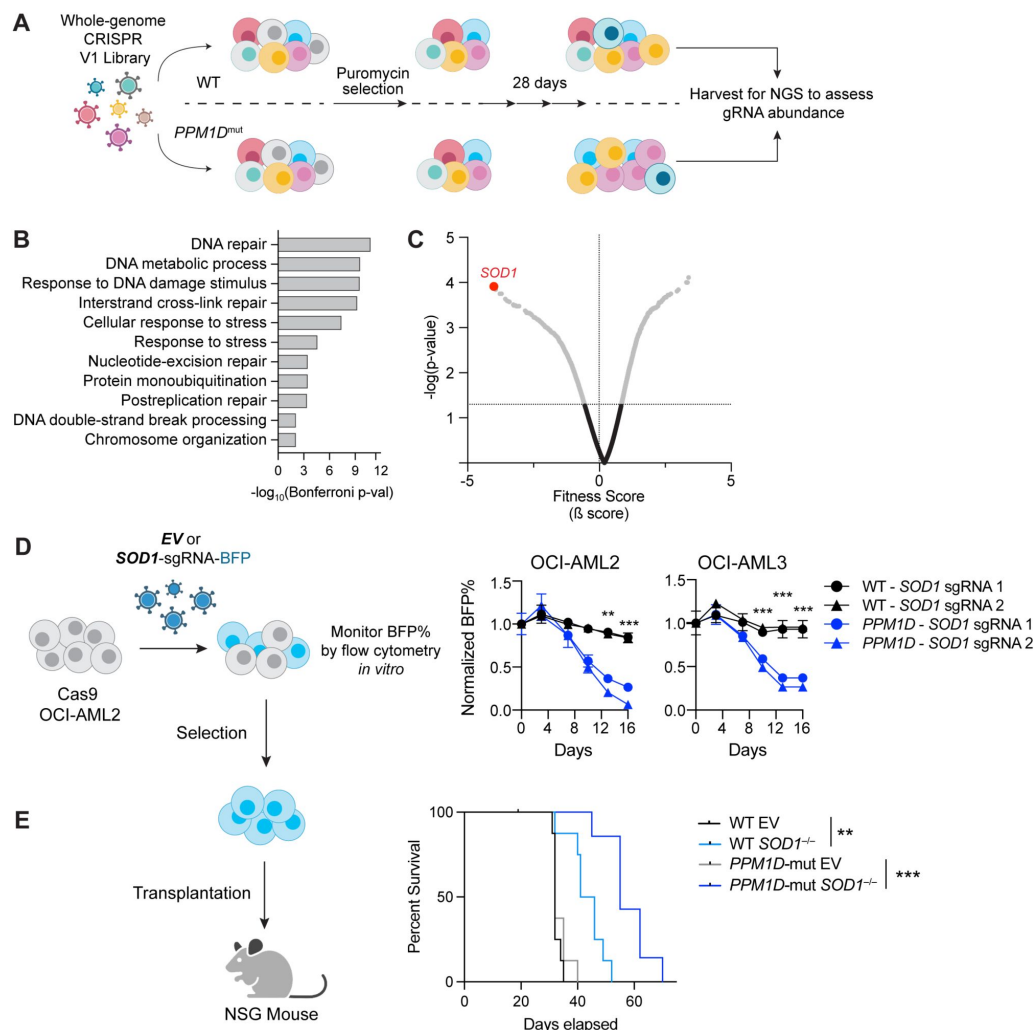


Figure 1.

***SOD1* is a synthetic lethal vulnerability of *PPM1D*-mutant leukemia cells.**

(A) Schematic of whole-genome CRISPR dropout screen. WT Cas9-expressing OCI-AML2 and two isogenic *PPM1D*-mutant lines were transduced with the Human Improved Whole Genome Knockout CRISPR library V1 containing 90,709 guide RNAs (gRNAs) targeting 18,010 human genes at low multiplicity of infection (MOI~0.3). Each condition was performed in technical triplicates. Three days post-transduction, cells underwent puromycin selection for three days. Cells were harvested at day 10 as the initial timepoint and then harvested every three days afterwards. sgRNA sequencing was performed on cells collected on day 28. **(B)** Top biological processes based on gene ontology analysis of the top 37 genes essential for *PPM1D*-mutant cell survival. Enrichment and depletion of guides and genes were analyzed using MAGeCK-VISPR by comparing read counts from each *PPM1D*-mutant cell line replicate with counts from the initial starting population at day ten. **(C)** Volcano plot of synthetic lethal hits ranked by fitness score with a negative score indicating genes for which their knockout leads to decreased growth or survival. *SOD1* (highlighted) was the top hit from the screen. **(D)** Left: Schematic of competitive proliferation assays used for validation of CRISPR targets. Right: WT and *PPM1D*-mutant Cas9-OCI-AML2 and Cas9-OCI-AML3 cells were transduced with lentiviruses containing a single *SOD1*-gRNA with a blue fluorescent protein (BFP) reporter. Cells were assayed by flow cytometry every 3-4 days and normalized to the BFP percentage at day 3 post-transduction. Two unique gRNAs against *SOD1* were used per cell line and each condition was performed in technical duplicates; multiple unpaired t-tests, ** $p < 0.01$, *** $p < 0.001$. **(E)** Left: Cas9-expressing WT and *PPM1D*-mutant cells were transduced with control or sg*SOD1*-containing lentiviruses and underwent puromycin (3 $\mu\text{g/mL}$) selection for three days prior to transplantation. Sublethally-irradiated (250 cGy) NSG mice were intravenously transplanted with 3×10^6 cells. Right: Kaplan-Meier survival curve of mice transplanted with WT or *PPM1D*-mutant (grey) leukemia cells with or without *SOD1*-deletion. The median survival of mice transplanted with WT, WT/*SOD1*^{-/-}, *PPM1D*^{mut}, and *PPM1D*^{mut}/*SOD1*^{-/-} leukemia cells was 32, 43, 32, and 55 days, respectively; Mantel-Cox test, ** $p < 0.01$, *** $p < 0.001$.

to the availability of SOD1 small molecule inhibitors that are being tested in clinical trials (Lin et al., 2013 [↗](#); Lowndes et al., 2008 [↗](#)). Therefore, we decided to further investigate the role of SOD1 in promoting *PPM1D*-mutant cell survival.

To validate the essentiality of SOD1 in *PPM1D*-mutant cells, we performed *in vitro* competitive proliferation assays in two different acute myeloid leukemia (AML) cell lines, OCI-AML2 and OCI-AML3. We transduced isogenic WT and *PPM1D*-mutant Cas9-expressing cells with either empty vector (EV) or sgSOD1-expressing lentiviral vectors containing a blue-fluorescent protein (BFP) reporter. We validated the loss of SOD1 protein expression by western blot (**Figure 1 [↗](#) – figure supplement 1D**) and confirmed that transduction of the empty vector control did not alter cellular fitness (**Figure 1 [↗](#) – figure supplement 1E**). While loss of SOD1 had minimal effects on the fitness of WT cells, *PPM1D*-mutant cells with SOD1 deletion had significant reduction in cellular growth in both OCI-AML2 and OCI-AML3 cells *in vitro* (**Figure 1D [↗](#)**).

To test if SOD1-deletion affected the fitness of *PPM1D*^{mut} vs WT leukemia cells *in vivo*, we transplanted *PPM1D*-mutant and -WT OCI-AML2 cells with or without SOD1 deletion into immunodeficient (NSG) mice. Mice transplanted with control *PPM1D*-mutant and -WT cells (with intact SOD1) had a similar median survival of 32 days. When SOD1 was deleted, the survival of mice transplanted with *PPM1D*-WT leukemia cells increased to a median of 43 days. Importantly, the survival of mice transplanted with *PPM1D*^{mut}-SOD1^{-/-} cells was even more significantly extended to a median time of 55 days (**Figure 1E [↗](#)**). These data provide an *in vivo* validation of the CRISPR screen demonstrating a differential dependency between *PPM1D*-mutant vs -WT cells on SOD1. Broadly, these results show that loss of SOD1 confers a disadvantage to leukemia cells that is markedly amplified in the context of the *PPM1D* truncating mutation.

***PPM1D*-mutant cells are sensitive to SOD1 inhibition and have increased oxidative stress**

We next wanted to test if pharmacologic inhibition of SOD1 could mimic the genetic deletion of SOD1. We used two different SOD1 inhibitors, 4,5-dichloro-2-m-tolyl pyridazin-3(2H)-one (also known as lung cancer screen-1, or LCS-1) and Bis-choline tetrathiomolybdate (ATN-224), which work by different mechanisms. LCS-1 is a small molecule that binds to SOD1 and disrupts its activity (Somwar et al., 2011 [↗](#)), while ATN-224 is a copper chelator that reduces SOD1 activity by decreasing the availability of copper ions, which are an essential SOD1 cofactor (Juarez et al., 2006 [↗](#)).

To study the sensitivity of the mutant cells to SOD1 inhibition, we engineered truncating *PPM1D* mutations into three patient-derived AML cell lines, MOLM-13, OCI-AML2, and OCI-AML3, which harbor distinct genetic backgrounds and AML driver mutations. At baseline, we found that *PPM1D*-mutant cells had increased SOD activity compared to WT cells and confirmed that SOD activity was significantly inhibited upon treatment with ATN-224 in a dose-dependent manner (**Figure 2-figure supplement 1A [↗](#)**). In addition, ATN-224 induced a significantly greater proportion of apoptotic *PPM1D*-mutant than *PPM1D*-WT cells (**Figure 2-figure supplement 1B [↗](#)**). *PPM1D*-truncating mutations conferred significant sensitivity to SOD1 inhibition compared to their WT counterparts in all three AML cell lines (**Figure 2A [↗](#), Figure 2-figure supplement 2A [↗](#)**). To determine if this cytotoxicity was dependent on oxidative stress, we treated the cells with SOD1 inhibitors in combination with an antioxidant, N-acetylcysteine (NAC). Importantly, NAC supplementation was able to completely rescue the sensitivity of mutant cells to both LCS-1 and ATN-224 treatment (**Figure 2B [↗](#), Figure 2-figure supplement 1C [↗](#)**), suggesting that ROS generation contributes to the sensitivity of mutant cells to SOD1 inhibition.

Activating mutations in oncogenes often lead to increased ROS generation by altering cellular metabolism, inducing replication stress, or dysregulating redox homeostasis (Maya-Mendoza et al., 2015 [↗](#); Park et al., 2014 [↗](#)). We therefore hypothesized that *PPM1D*-mutant cells have increased

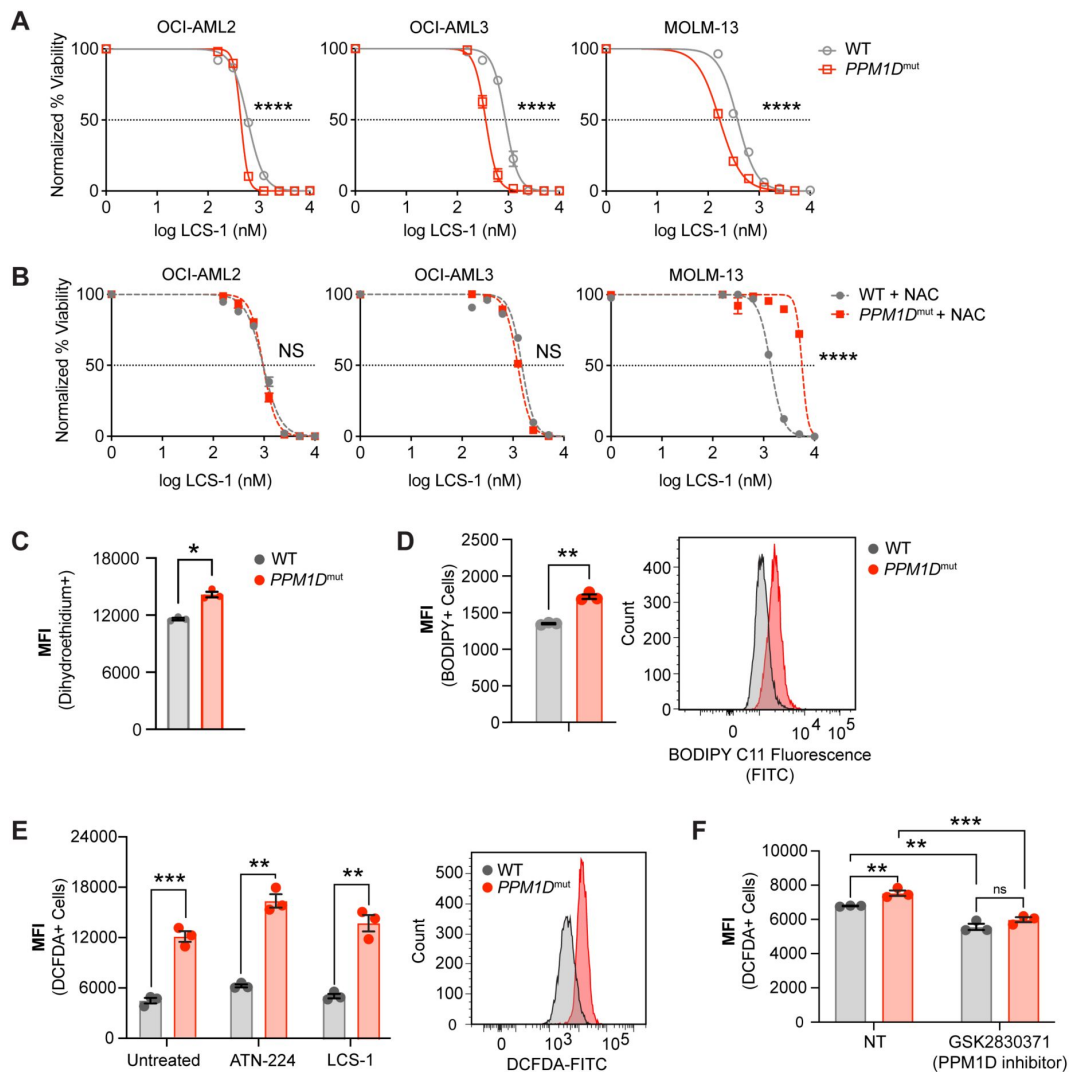


Figure 2.

PPM1D-mutant cells are sensitive to SOD1 inhibition and have increased oxidative stress.

(A,B) Dose response curves for cell viability with SOD1-inhibitor (LCS-1) (A) or LCS-1 in combination with 0.25 μ M NAC (B) in WT and PPM1D-mutant leukemia cell lines after 24-hours. Mean $\pm$ SD (n=3) is shown with a non-linear regression curve. All values are normalized to the baseline cell viability with vehicle, as measured by MTT assay. (C) Endogenous cytoplasmic superoxide levels of WT and PPM1D-mutant leukemia cell lines were measured using dihydroethidium (5 μ M). The mean fluorescence intensity (MFI) of dihydroethidium was measured by flow cytometry. Mean $\pm$ SD (n=3) is shown. (D) Lipid peroxidation measured using BODIPY 581/591 staining (2.5 μ M) of WT and PPM1D-mutant OCI-AML2 cells. The MFI was measured by flow cytometry. Mean $\pm$ SD (n=3) is shown. (E-F) Measure of total reactive oxygen species using 2',7'-dichlorofluorescein diacetate (DCFDA) staining (10 μ M) measured by flow cytometry. WT and PPM1D-mutant OCI-AML2 cells were measured at baseline and 24-hrs after SOD1 inhibition (ATN-224 12.5 μ M, LCS-1 0.625 μ M) (E) or 24-hrs after pharmacologic PPM1D inhibition (GSK2830371, 5 μ M) (F); unpaired t-tests were used for statistical analyses, ns=non-significant (p>0.05), **p<0.01, ***p<0.001, ****p<0.0001.

oxidative stress, leading to reliance on SOD1 for protection. SOD1 catalyzes the breakdown of superoxide into hydrogen peroxide and water. Therefore, we assessed cytoplasmic and mitochondrial superoxide levels using dihydroethidium and Mitosox Green, respectively. These fluorogenic dyes are rapidly oxidized by superoxide, but not other types of ROS, to produce green fluorescence. We observed that in the absence of exogenous stressors, *PPM1D*-mutant cells had a moderate increase in superoxide radicals (**Figure 2C** [↗](#), **Figure 2-figure supplement 1C** [↗](#)). SOD2 is the primary superoxide dismutase in the mitochondria responsible for catalyzing superoxide into H_2O_2 . Given the increase in mitochondrial superoxide levels, we assessed levels of SOD2 protein levels. Surprisingly, there were no baseline differences or compensatory changes in SOD2 after *SOD1* deletion (**Figure 2-figure supplement 2C** [↗](#)).

Free radicals can be detrimental to cells due to their ability to oxidize proteins, lipids, and DNA. Therefore, we also measured levels of lipid peroxidation as an additional measure of oxidative stress. Consistent with the increase in superoxide radicals, we observed a concurrent increase in lipid peroxidation in the *PPM1D*-mutant cells (**Figure 2D** [↗](#)). Using 2',7'-dichlorofluorescein diacetate (DCFDA) staining to measure total ROS levels, we observed that *PPM1D*-mutant cells harbored more total ROS compared to WT cells (**Figure 2E** [↗](#)).

To investigate whether the observed elevated ROS was a characteristic of other *PPM1D*-mutant cell lines, we measured ROS levels in two different germline models. Humans with germline mutations in *PPM1D* were first described by Jansen et al. in 2017 in patients with intellectual disability. This neurodevelopmental condition is named Jansen-de Vries syndrome (JdVS, OMIM #617450) and is characterized by frameshift or nonsense mutations in the last or second-to-last exons of the *PPM1D* gene. These mutations result in functionally active, truncated mutant proteins like those exhibited in human cancers and clonal hematopoiesis. Lymphoblastoid cell lines (LCLs) were generated from these JdVS patients by Jansen et al. (Jansen et al., 2017 [↗](#); Wojcik et al., 2023 [↗](#)).

In addition to human *PPM1D*-mutant LCLs, we also generated mouse embryonic fibroblasts (MEFs) from a germline mouse model harboring a heterozygous truncating mutation in the terminal exon of *Ppm1d* (Hsu et al., 2018 [↗](#)). When we measured total ROS from both the JdVS LCLs and the *Ppm1d*-mutant MEFs compared to their WT counterparts, both mutant models exhibited greater levels of total ROS (**Figure 2-figure supplement 2D** [↗](#), **2E**). Additionally, *PPM1D*-mutant LCLs were also more sensitive to pharmacologic SOD1 inhibition compared to the WT LCL line, GM12878 (**Figure 2-figure supplement 2F** [↗](#)). These results demonstrate that *PPM1D* mutations not only increase ROS in the context of cancer, where cellular metabolism is often altered, but can also alter redox homeostasis in non-transformed cells.

To determine if mutant *PPM1D* was associated with ROS generation, we treated isogenic OCI-AML2 WT and *PPM1D*-mutant cells with a *PPM1D* inhibitor, GSK2830371, for 24 hours. We found that pharmacologic inhibition of *PPM1D* mildly decreased ROS levels in both WT and *PPM1D* mutant cells (**Figure 2F** [↗](#)), suggesting a link between *PPM1D* and ROS production. Altogether, these data demonstrate that SOD1 inhibition leads to cytotoxicity in the mutant cells due to increased oxidative stress induced by mutant *PPM1D*.

***PPM1D*-mutant leukemia cells have altered mitochondrial function**

Mitochondria are the primary source of ROS within the cell, as the electron transport chain is a major site of ROS production during oxidative phosphorylation. We next asked whether the observed increase in ROS in *PPM1D*-mutant cells was due to differences in mitochondrial abundance. We used two independent methods to measure mitochondrial mass, including MitoTracker Green flow cytometry (**Figure 3A** [↗](#)) and western blot analysis of mitochondrial complex proteins (**Figure 3B** [↗](#)). However, we did not observe a difference in mitochondrial mass with either method. This finding suggests that mechanisms other than a change in mitochondrial abundance are responsible for the increase in ROS levels in mutant cells, such as alterations in mitochondrial metabolism or changes in ROS scavenging systems.

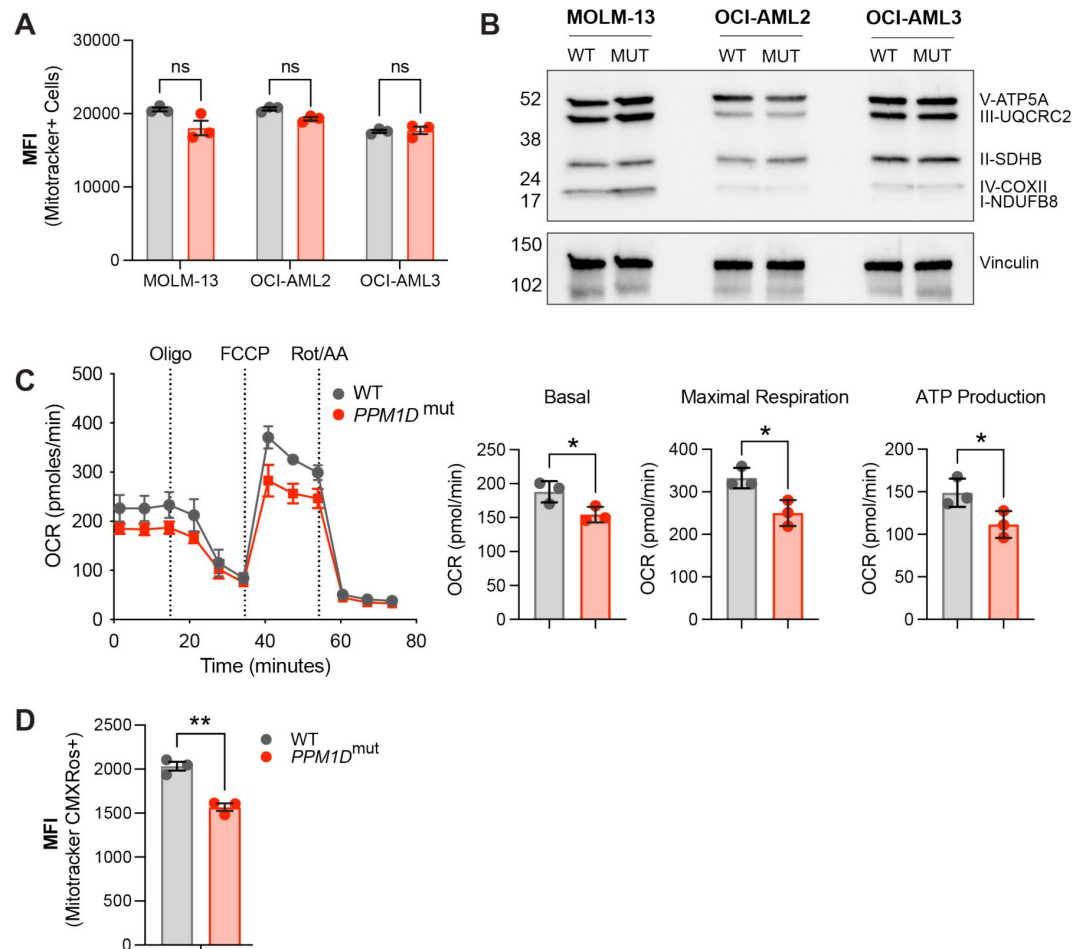


Figure 3.

***PPM1D*-mutant cells have altered mitochondrial function.**

(A) Mitochondrial mass of WT and *PPM1D*-mutant leukemia cells was determined using MitoTracker Green (100 nM) and the mean fluorescence intensity was analyzed by flow cytometry. Data represents mean $\pm$ SD of triplicates. At least three independent experiments were conducted with similar findings; unpaired t-tests. **(B)** Immunoblot of WT and *PPM1D*-mutant cell lysates probed with the human OXPHOS antibody cocktail (1:1000) and vinculin (1:2000). **(C)** Measurement of mitochondrial oxygen consumption rate (OCR) by seahorse assay in WT and *PPM1D*-mutant OCI-AML2 cells after treatment with oligomycin (1.5 μ M), FCCP (0.5 μ M), and rot/AA (0.5 μ M). Quantification of basal, maximal, and ATP-linked respiration are shown. Data shown are the mean $\pm$ SD of technical triplicates. **(D)** Mitochondrial membrane potential of WT and *PPM1D*-mutant OCI-AML2 cells was measured using MitoTracker CMXRos (400 nM). The mean fluorescence intensity (MFI) was measured and analyzed by flow cytometry. Data represents mean $\pm$ SD of triplicates, unpaired t-test, ns=non-significant ($p>0.05$), * $p<0.05$, ** $p<0.01$.

To assess mitochondrial function, we performed seahorse assays in WT and *PPM1D*-mutant cells. Our seahorse assays revealed that the mutant cells have decreased mitochondrial respiration, as indicated by decreased basal, maximal, and ATP-linked respiration (**Figure 3C** [↗](#)). While *PPM1D*-mutant MOLM-13 and OCI-AML3 cells also had decreased basal respiration, there were variable differences in maximal and ATP-linked respiration compared to WT, suggesting possible cell line differences affecting mitochondrial respiration (**Figure 3-figure supplement 1A** [↗](#), **B**). In addition to analyzing respiratory capacity, we also examined mitochondrial membrane potential (MMP) using the fluorescent dye Mitotracker CMXRos, which accumulates in the mitochondria in an MMP-dependent manner. We stained both WT and mutant cells with Mitotracker CMXRos and observed a decrease in MMP in the mutant cells (**Figure 3D** [↗](#)). Tracking cell numbers between the WT and mutant cell lines over time established this decrease in MMP was not due to altered cellular growth rates (**Figure 3-figure supplement 1C** [↗](#)). Altogether, these findings, along with decreased respiratory capacity and increased mitochondrial ROS, indicate a mitochondrial defect in *PPM1D*-mutant cells.

***PPM1D*-mutant cells have a reduced oxidative stress response**

Mitochondrial dysfunction and increased ROS production are closely intertwined. On one hand, mitochondrial dysfunction leads to increased ROS production as a result of impaired oxidative phosphorylation and increased electron leakage (Turrens, 2003 [↗](#)). On the other hand, sustained oxidative stress can directly damage mitochondrial components and mtDNA and compromise their function (Wallace, 2005 [↗](#)). To better understand the molecular basis for the observed mitochondrial dysfunction and dependency on SOD1, we performed bulk RNA-sequencing (RNA-seq) on Cas9-expressing WT and *PPM1D*-mutant OCI-AML2 cells transduced with *SOD1*-sgRNA to induce *SOD1* deletion or the empty vector (EV) control (**Figure 4-figure supplement 1A** [↗](#)). Both EV and *SOD1*-sgRNA vectors were tagged with a BFP reporter to identify transduced cells. The cells were collected ten-days post-transduction, the timepoint at which we observed 50% reduction of the *SOD1*-deletion cells during the *in vitro* proliferation assays, reasoning this would capture the effects of *SOD1*-deletion on cellular and metabolic processes while avoiding excessive cell death.

Analysis of the RNA-seq data revealed 2239 differentially expressed genes, with 1338 downregulated genes and 901 upregulated genes in the mutant cells compared to WT cells at baseline (**Figure 4-source data 1**). Gene set enrichment analysis (GSEA) of the differentially expressed genes showed an upregulation in genes related to cell cycle (GO: 0007049), cell division (GO: 0051301), DNA replication (GO: 005513), and mitophagy (GO: 0000423) in the *PPM1D*-mutant cells (**Figure 4A** [↗](#)). Interestingly, there was a significant downregulation of pathways related to the regulation of the oxidative stress response (GO: 1902882, **Figure 4-figure supplement 1B** [↗](#)), ROS metabolic processes (GO: 0072593), and oxidation reduction (GO: 0055114). Following *SOD1* deletion, the WT cells displayed notable upregulation of pathways associated with cell cycling, chromosome organization, cell division, and DNA repair. In contrast, the mutant cells showed significant downregulation of these same pathways (**Figure 4-figure supplement 1C** [↗](#)). Intriguingly, upon *SOD1* deletion, the mutant cells exhibited an upregulation in the response to oxidative stress (GO:0006979, **Figure 4-figure supplement 1D** [↗](#)). This finding suggests a reactive transcriptional response to the heightened ROS levels resulting from the loss of SOD1.

As *PPM1D* is a phosphatase that can directly modulate the activation state of proteins, we examined whether there were alterations in protein and phosphoprotein levels in *PPM1D*-mutant cells using reverse phase protein array (RPPA) analysis, mirroring the experimental design used for bulk RNA-seq (**Figure 4-figure supplement 1A** [↗](#)). By focusing on differential protein expression between wild-type (WT) and *PPM1D*-mutant cells, we aimed to capture the post-translational regulatory events that could contribute to the mitochondrial dysfunction observed in the mutants. The RPPA analysis of over 200 (phospho-)proteins covering major signaling pathways identified 128 differentially expressed proteins between *PPM1D*-mutant and control WT OCI-AML2 cells (a panel of 264 proteins), with 67 downregulated proteins and 61 upregulated proteins (**Figure 4-figure supplement 2A** [↗](#), **source data 2**). Notably, over-representation analysis showed

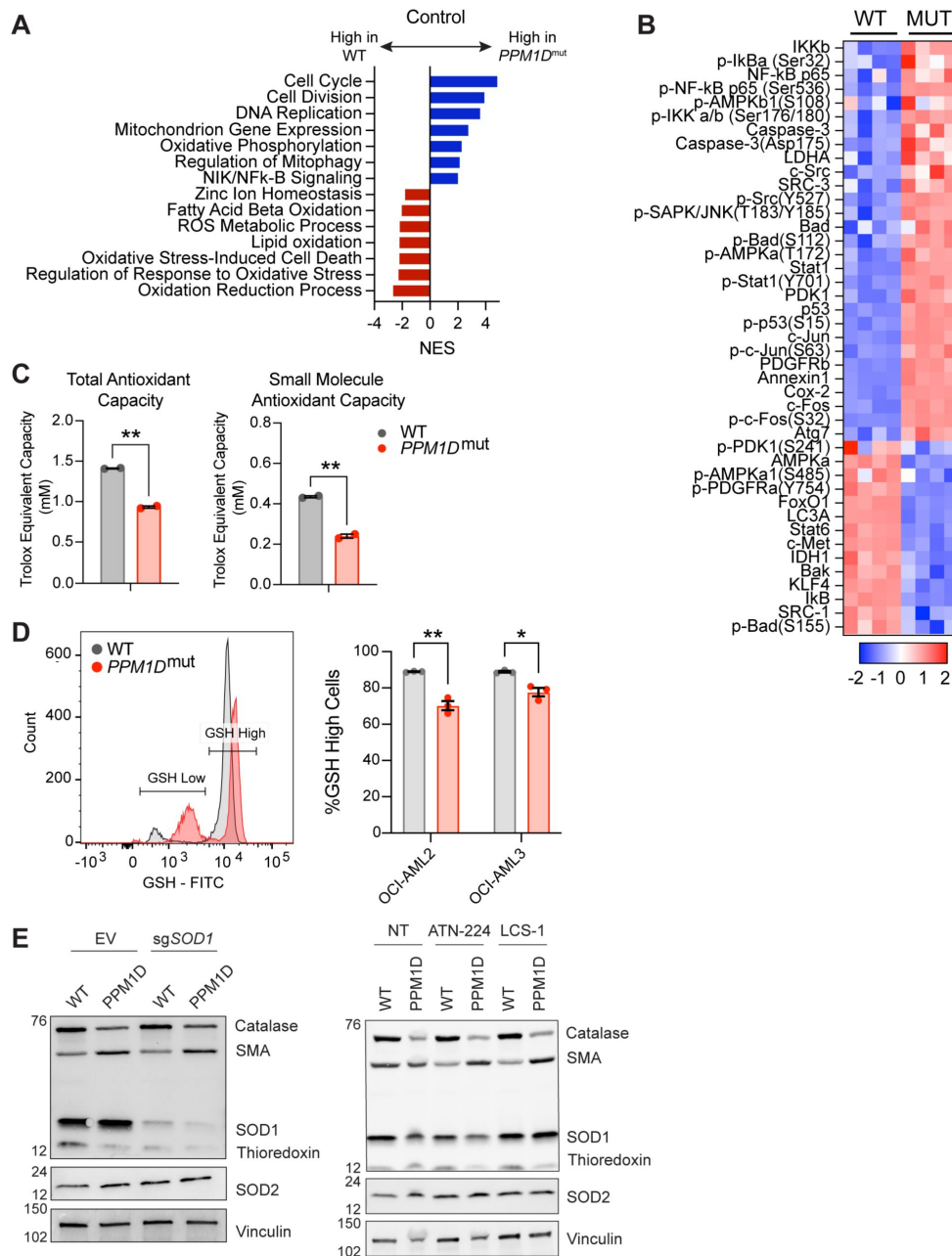


Figure 4.

PPM1D-mutant cells have a reduced oxidative stress response.

(A) RNA-seq GSEA analysis of *PPM1D*-mutant cells compared to WT Cas9-OCI-AML2 cells. Significantly up- and downregulated pathways are indicated by the blue and red bars, respectively. Normalized enrichment scores (NES) are shown with FDR < 0.25. **(B)** RPPA profiling of WT and *PPM1D*-mutant OCI-AML2 cells. Proteins from the "Response to Oxidative Stress" pathway have been selected for the heatmap. Each column represents a technical replicate. See Figure 4-source data 2 for the raw data. **(C)** Total- and small molecule antioxidant capacity of WT and *PPM1D*-mutant cells performed in technical duplicates. **(D)** Intracellular glutathione (GSH) levels measured by flow cytometry using the Intracellular GSH Detection kit (abcam). Left: representative flow cytometry plot demonstrating the gating for GSH-high and GSH-low populations. Right: quantification of the percentage of GSH-high cells for each cell line. Mean $\pm$ SEM (n=3) are shown. **(E)** Immunoblot of WT and *PPM1D*-mutant OCI-AML2 after transduction with the empty vector (EV) control and after *SOD1* deletion (left) or after treatment with *SOD1* inhibitors for 16 hours (right, ATN-224 12.5 μ M, LCS-1 1.25 μ M). Lysates were probed with an anti-oxidative stress defense cocktail (1:250), *SOD2* (1:1000), and vinculin (1:2000). SMA=smooth muscle actin. Student t-tests were used for statistical analysis; **p<0.01, *p<0.05.

that among the differentially expressed proteins, there was a significant enrichment in the “Response to Oxidative Stress” pathway in the mutant cells ($-\log_{10}(\text{pValue}) = 24.164$) compared to WT, with a particular emphasis on the downregulated proteins of this pathway ($-\log_{10}(\text{pValue}) = 15.457$, **Figure 4B** [↗](#), **Figure 4—source data 3**). While the RNA-seq suggested a transcriptional upregulation of the response to oxidative stress in the mutant cells after *SOD1* deletion, the RPPA data revealed that the mutant cells continued to exhibit decreased expression in proteins associated with the oxidative stress response (**Figure 4—figure supplement 2B** [↗](#)). Taken together, these findings suggest that *PPM1D*-mutant cells have an inherent impairment in their baseline response to oxidative stress.

To further explore the diminished oxidative stress response in the mutant cells, we assessed their total- and small-molecule-antioxidant capacity. Total antioxidant capacity refers to the overall ability of the cells to counteract free radicals and reduce oxidative damage. This includes enzymatic antioxidants such as catalase, SODs, and peroxidases. Small molecule antioxidant capacity measures the capacity of low molecular weight antioxidants, such as glutathione and vitamin E, to neutralize ROS (Hawash et al., 2022 [↗](#)). Our results showed that *PPM1D*-mutant cells have significantly reduced total and small-molecule antioxidant capacity compared to WT cells (**Figure 4C** [↗](#)).

Subsequently, we measured intracellular glutathione (GSH), a pivotal antioxidant crucial for maintaining cellular redox balance and protecting against oxidative stress. Strikingly, our analysis revealed a higher proportion of mutant cells with diminished GSH levels compared to their WT counterparts (**Figure 4D** [↗](#)). We also measured the protein levels of key antioxidant enzymes by western blot. While we saw similar protein levels of SOD1 in both WT and mutant cells, we observed a reduction in the thioredoxin and catalase levels (**Figure 4E** [↗](#)). These results provide evidence to support the RNA-seq and RPPA findings that *PPM1D*-mutant cells have impaired antioxidant defense mechanisms, leading to an elevation in ROS levels and reliance on SOD1 for protection.

***PPM1D* mutations increase genomic instability and impair non-homologous end-joining repair**

In addition to a decreased response to oxidative stress, the RNA-seq GSEA analysis also revealed differential responses to DNA repair. Upon *SOD1* deletion, WT cells significantly upregulated the regulation of DNA repair (GO:0006281), double-stranded break repair (GO:0006302), homologous recombination (GO:0035825), and more. However, there was a striking downregulation of DNA repair pathways after deletion of *SOD1* in the mutant cells (**Figure 4—figure supplement 1C** [↗](#)). *PPM1D* plays a key role in suppressing the DNA damage response (DDR) by dephosphorylating, thereby inactivating, p53 and other key upstream and downstream effectors of the pathway. Truncating mutations and amplifications in *PPM1D* that lead to increased *PPM1D* activity may therefore inhibit DNA damage repair and increase genomic instability. Oxidative stress and ROS also pose endogenous challenges to genomic integrity. Therefore, we hypothesized that due to the increase in ROS within the mutant cells, loss of *SOD1* may lead to unsustainable accumulation of DNA damage and overwhelm the mutant cell’s DNA repair capacity.

To test this hypothesis, we first sought to establish the baseline levels of DNA damage in *PPM1D*-altered cells. We performed alkaline comet assays in mouse embryonic fibroblasts and found a significant increase in single- and double-stranded DNA breaks in mutant cells compared to WT (**Figure 5A** [↗](#)). As ROS are known to contribute to oxidative DNA damage, we further assessed the levels of 8-oxo-2'-deoxyguanosine (8-oxo-dG), a well-established marker of oxidative DNA damage. Strikingly, the mutant cells demonstrated elevated levels of oxidative DNA damage at baseline (**Figure 5B** [↗](#)). We also performed metaphase spreads in mouse primary B-cells to investigate chromosomal aberrations, which are consequences of abnormal double-stranded break repair. WT and *Ppm1d*-mutant mouse primary resting CD43⁺ B-cells were purified from spleens and

stimulated with LPS, IL-4, and CD180 to induce proliferation. The cells were then treated with either low- or high-dose cisplatin for 16-hours. Consistent with our comet assay findings, we observed that *Ppm1d*-mutant cells harbored approximately two-fold more chromosomal breaks per metaphase after exposure to cisplatin (**Figure 5C**). When we classified the chromosomal aberrations into subtypes, we observed that the mutant cells had increased numbers of each type of aberration. These results demonstrate that mutations in *PPM1D* increase genomic instability.

To further assess the DNA repair efficiency of *PPM1D*-mutant cells, we utilized U2OS DNA repair reporter cell lines which express a green fluorescent protein (GFP) cassette when specific DNA repair pathways are active after stimulation when the I-SceI restriction enzyme is induced to stimulate a double-stranded break (DSB). To test for homologous recombination (HR), tandem defective GFP genes can undergo HR to generate GFP+ cells. Non-homologous end joining (NHEJ) repairs a defective GFP in a distinct cassette (Weinstock et al., 2006). Because the U2OS parental line harbors an endogenous heterozygous *PPM1D* truncating mutation (R458X) (Kleiblova et al., 2013), we corrected the lines to generate the isogenic *PPM1D* WT control (**Figure 5-figure supplement 1A**).

With two isogenic clones for each reporter cell line, we transfected the *PPM1D*-WT and -mutant U2OS clones with I-SceI and measured GFP expression by flow cytometry after 48 hours. Our results showed similar levels of HR-mediated repair in both WT and mutant clones (**Figure 5D**). Prior studies have shown that WT *PPM1D* promotes HR by forming a stable complex with BRCA1-BARD1, thereby enhancing their recruitment to DSB sites (Burdova et al., 2019). Although gain-of-function mutations in *PPM1D* lead to persistent *PPM1D* activity, it may not necessarily result in increased HR repair. Several factors can limit the extent of HR enhancement. For instance, HR is typically restricted to the S/G2 phase of the cell cycle and is a multi-step process that begins with DNA end resection (Xu and Xu, 2020). This is a crucial initial step that generates single-stranded DNA overhangs to facilitate strand invasion and recombination (Grugge and Symington, 2021). Therefore, the impact of mutant *PPM1D* on HR may be constrained by the efficiency of DNA end resection and cell cycling, among other regulatory mechanisms within the HR pathway.

In contrast, we saw significantly decreased NHEJ repair in the *PPM1D*-mutant clones (**Figure 5E**). This downregulation of NHEJ may be due to diminished activation of γH2AX and ATM. These two proteins serve as key upstream regulators within the DDR and are subject to dephosphorylation by *PPM1D* (Cha et al., 2010; Lu et al., 2005). In addition, prior studies have also shown that *PPM1D* modulates lysine-specific demethylase 1 (LSD1) activity, which is important for facilitating the recruitment of 53BP1 to DNA damage sites through RNF168-dependent ubiquitination (Peng et al., 2015). *PPM1D* mutations may therefore lead to impairment of NHEJ through dysregulation of 53BP1 recruitment. To confirm this, we performed immunofluorescence imaging of Rad51 and 53BP1 foci. The recruitment of Rad51 and 53BP1 to the sites of DNA damage are important for the activation of HR and NHEJ, respectively. We analyzed mouse embryonic fibroblasts at baseline and after irradiation (10 Gy) and observed similar numbers of Rad51 foci in *Ppm1d*-mutant and WT cells (**Figure 5-figure supplement 1B**). In contrast, *Ppm1d*-mutant MEFs had fewer 53BP1 foci, indicating decreased NHEJ repair capacity that was consistent with our U2OS reporter line findings (**Figure 5-figure supplement 1C**). Comet assays were performed in parallel with the immunofluorescence experiments to show that the mutant cells had increased DNA damage (**Figure 5-figure supplement 1D**). Therefore, the decrease in foci was not due to resolution of DNA damage, but rather due to inefficient DNA repair.

In light of the elevated levels of DNA damage and compromised DNA repair observed in the *PPM1D*-mutant cells, we hypothesized that loss of *SOD1* may exacerbate genomic instability, ultimately leading to mutant cell death. To assess this hypothesis, we performed comet assays after *SOD1* deletion. Contrary to our hypothesis, genetic deletion of *SOD1* did not result in a significant increase in DNA breaks in neither WT nor mutant cells (**Figure 5F**). These results suggest that

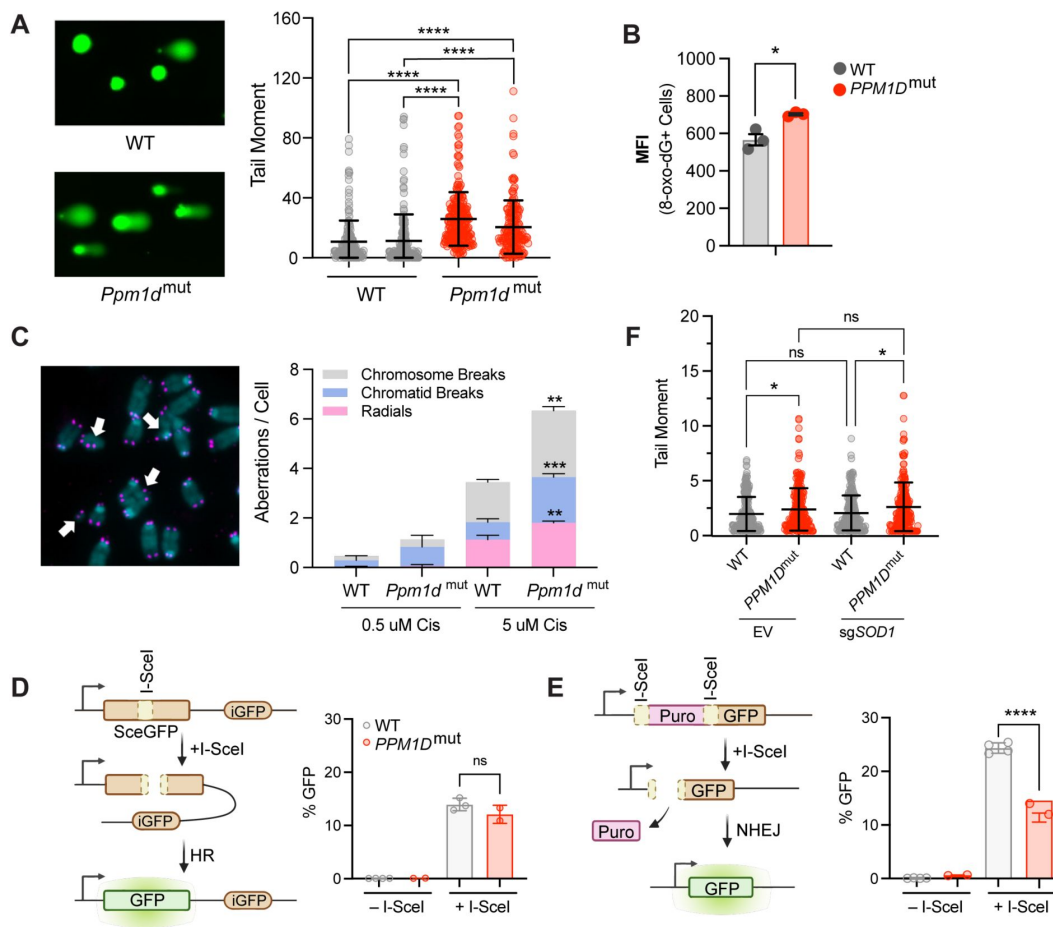


Figure 5.

***PPM1D* mutations increase genomic instability and impair non-homologous end-joining.**

(A) Left: Representative images of comet assays of mouse embryonic fibroblasts (MEFs). Two biological replicates were assessed for each genotype. Right: Quantification of $n \geq 150$ comets per experimental group with the Comet IV software; 2way ANOVA. ****
(B) Mean fluorescent intensity (MFI) of 8-oxo-dG lesions within WT and *PPM1D*-mutant OCI-AML2 cells as measured by flow cytometry; student's t-test. *
(C) Left: Representative images of metaphase spreads of WT and *Ppm1d*-mutant mouse primary B-cells treated with low (0.5 μ M) or high (5 μ M) doses of cisplatin. Right: $n \geq 50$ metaphase cells were quantified in each experimental condition for chromosomal aberrations (white arrows). $n=2$ biological replicates used for each genotype. Student's t-test was used for statistical analysis. **
(D-E) Left: Schematic of the homologous recombination **(D)** or non-homologous end-joining **(E)** U2OS DNA damage repair cassettes. Right: Quantification of GFP% analyzed by flow cytometry 48-hours after induction of DNA damage by I-SceI transduction; student's t-test. ns
(F) Comet assay quantification of WT and *PPM1D*-mutant Cas9-OCI-AML2 cells six days after lentiviral transduction with the empty vector (EV) control, or *sgSOD1* to induce *SOD1* deletion. Quantification and analyses of tail moments were performed using the Comet IV software. $n \geq 150$ comets were scored per experimental group; 2way ANOVA. Data are mean $\pm$ SD ($n=3$), ns=non-significant ($p > 0.05$), * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

the vulnerability of *PPM1D*-mutant cells to SOD1 loss is not necessarily mediated by an exacerbation of DNA damage but rather by other consequences of oxidative stress and ROS imbalance.

Discussion

The search for synthetic-lethal strategies for cancer therapy has gained significant attention in recent years due to the potential to identify new therapeutic targets that exploit tumor-specific vulnerabilities. In this study, we performed whole-genome CRISPR/Cas9 screening to uncover synthetic-lethal partners of *PPM1D*-mutant leukemia cells. Our screen revealed that *SOD1* was the top essential gene for *PPM1D*-mutant cell survival, a dependency that was validated *in vivo*. Ongoing efforts are underway to develop SOD1 inhibitors for the treatment of cancer and ALS (Abati et al., 2020 [↗](#); Huang et al., 2000 [↗](#)), and it is conceivable these may be useful in the context of *PPM1D* mutation.

To explore this concept, we tested the sensitivity of WT and *PPM1D*-mutant cells to known SOD1 inhibitors ATN-224 and LCS-1. We found that *PPM1D*-mutant cell lines were significantly more sensitive to these compounds compared to WT. This sensitivity could be rescued upon supplementation with the antioxidant, NAC, consistent with a role in reducing the impact of reactive oxygen species. However, given potential off-target effects of LCS-1 (Ling et al., 2022 [↗](#); Steverding and Barcelos, 2020 [↗](#)) we cannot verify that the cytotoxic effects are via its activity toward SOD1. Similarly, we cannot rule out that effects of ATN-224 are not due to other effects caused by copper chelation (Chidambaram et al., 1984 [↗](#); Lee et al., 2013 [↗](#); Lowndes et al., 2008 [↗](#); Lowndes et al., 2009 [↗](#)). Further work to determine the potential of SOD mimetics like TEMPOL and MnTBAP in mitigating the effects of SOD1 inhibition would be valuable in confirming the specificity of the inhibitors for our underlying phenotype.

We also investigated the mechanisms underlying the dependency on SOD1 and characterized the redox landscape of *PPM1D*-mutant cells, which revealed significant oxidative stress and mitochondrial dysfunction. Recent studies have suggested that PPM1D is indirectly associated with energy metabolism via dephosphorylation of the ataxia telangiectasia mutated (ATM) protein. ATM promotes mitochondrial homeostasis, and therefore sustained inactivation of ATM could lead to potential mitochondrial dysfunction (Bar et al., 2023 [↗](#); Guleria and Chandna, 2016 [↗](#); Valentin-Vega et al., 2012 [↗](#)). However, oxidative stress and mitochondrial dysfunction are closely related, and it is difficult to dissect the driving factor. We therefore performed RNA-sequencing and RPPA analysis to better understand the underlying processes contributing to the heightened oxidative stress observed in the mutant cells. Our analyses indicated a diminished response to oxidative stress in the mutant cells. These findings may suggest a self-amplifying cycle whereby dysregulation of ROS scavenging systems increases ROS levels, which in turn leads to mitochondrial dysfunction, which further exacerbates oxidative stress. Hence, the additional impairment of ROS detoxification mechanisms within the cell, such as the loss of SOD1, has detrimental consequences for the viability of mutant cells.

The loss of SOD1 leads to increased O_2^- levels and reduced intracellular H_2O_2 . These two ROS species play especially important roles as signaling messengers that control cellular proliferation, differentiation, stress responses, inflammatory responses, and more (Sauer et al., 2001 [↗](#); Sies and Jones, 2020 [↗](#); Thannickal and Fanburg, 2000 [↗](#)). These effects are mediated through the reversible oxidation and reduction of cysteine residues (Poole, 2015 [↗](#)) that have significant effects on key signaling proteins including Erk1/2, protein phosphatases, and more. Therefore, while ROS levels may be significantly impacted by the loss of SOD1, we cannot rule out the possibility of altered ROS-driven signaling, rather than ROS-induced damage, as an underlying mechanism for our results.

Multiple mechanisms may underlie the suppressed oxidative stress response observed in *PPM1D*-mutant cells. One possible explanation is through PPM1D-mediated inhibition of p53. p53 exhibits complex and context-dependent roles in cellular responses to oxidative stress, and its functions can vary depending on the severity of stress encountered by the cell (Kang et al., 2013; Liang et al., 2013; Sablina et al., 2005). Under mild or moderate oxidative stress conditions, p53 may protect the cell from ROS by inducing the transcription of genes such as superoxide dismutase (SOD), glutathione peroxidase (GPx), and others (Dhar et al., 2011; Peugot et al., 2014; Sablina et al., 2005; Tan et al., 1999). However, under severe or prolonged oxidative stress, the pro-apoptotic functions of p53 may promote ROS production to eliminate cells that have accumulated excessive DNA damage or irreparable cellular alterations. The duality of these anti- and pro-oxidant functions of p53 highlight its intricate role in modulating responses to oxidative stress. How PPM1D affects the switch between these functions of p53 is not understood. Furthermore, the extent to which the dependency on SOD1 observed in *PPM1D*-mutant cells is mediated through p53 remains unclear and requires deeper exploration to better understand the context in which SOD1 inhibitors can be used in cancer therapy.

Oxidative stress and DNA damage are intimately linked processes that frequently co-occur. Our study also investigated the interplay between PPM1D, DNA damage, and oxidative stress. We demonstrated significant genomic instability of *PPM1D*-mutant cells at baseline and further characterized the effects of mutant PPM1D on specific DNA repair pathways. While previous studies have suggested a role for PPM1D in modulating HR and NHEJ (Burdova et al., 2019; Peng et al., 2015), our study is the first to demonstrate impaired NHEJ in *PPM1D*-mutant cells. Additionally, our study corroborated previous research demonstrating the synthetic-lethal relationship of *SOD1* and other DNA damage genes such as *RAD54B*, *BLM*, and *CHEK2* (Sajesh et al., 2013; Sajesh and McManus, 2015). However, *SOD1*-deletion did not exacerbate DNA damage, suggesting that the vulnerability of *PPM1D*-mutant cells to *SOD1* loss cannot be explained by increased DNA damage and may be more likely due to consequences of oxidative stress. Recent studies have shown that ATN-224 can enhance the anti-tumor effects of cisplatin by increasing ROS, decreasing glutathione content, and increasing DNA damage (Li et al., 2022). These results highlight the potential for combinatorial therapies to achieve therapeutic synergism and underscores the intricate relationship between ROS and DNA damage.

Interestingly, our screen also uncovered sensitivity of *PPM1D*-mutant cells to dropout of genes in the Fanconi Anemia (FA) DNA repair pathway including *BRIP1* (*FANCF*), *FANCI*, *FANCA*, *SLX4* (*FANCP*), *UBE2T* (*FANCT*), and *C19orf40* (*FAAP24*). The FA pathway plays a crucial role in facilitating the repair of interstrand crosslinks (Ceccaldi et al., 2016; Kottemann and Smogorzewska, 2013). Outside of DNA repair and replication, there is a growing body of evidence demonstrating mitochondrial dysfunction and redox imbalance in FA-patient cells (Korkina et al., 1992). Several FA proteins are implicated in the maintenance of mitochondrial metabolism and mitophagy (Cappelli et al., 2017; Kumari et al., 2014; Pagano et al., 2013; Sumpter et al., 2016). Interestingly, a few studies have described a convergence in the FA pathway with *SOD1*. Early work by Nordenson in 1977 found protective roles for SOD and catalase against spontaneous chromosome breaks in cells from FA patients. Another study demonstrated mitochondrial dysfunction, high ROS levels, and impaired ROS detoxification mechanisms in FA-deficient cell lines (Kumari et al., 2014). Interestingly, *SOD1* expression increased in response to H₂O₂ treatment in FA-intact cells, but not FA-deficient cells. These findings underscore the critical role of the FA pathway in redox homeostasis by maintaining mitochondrial respiratory function and suppressing intracellular ROS production. Even more importantly, it demonstrates a convergence in the FA pathway with *SOD1*, providing further support for our CRISPR dropout screen results.

In summary, our investigation sheds light on the role of mutant PPM1D in modulating cellular responses to oxidative stress and DNA repair in leukemia cells, offering valuable insights into the underlying molecular mechanisms. This research not only enhances our understanding of PPM1D-

mediated cellular responses, but also identifies potential therapeutic targets against *PPM1D*-mutant leukemia cells. However, it is important to acknowledge the limitations of our study. We recognize that while *PPM1D* mutations are frequently observed in patients with t-MN, they are rare in *de novo* AML (Hsu et al., 2018 [DOI](#)). While there is ample evidence that *PPM1D* is an oncogenic driver in many types of cancers (Ali et al., 2012 [DOI](#); Khadka et al., 2022 [DOI](#); Li et al., 2002 [DOI](#); Nguyen et al., 2010 [DOI](#); Wu et al., 2016 [DOI](#)), the clinical importance of targeting pre-malignant *PPM1D*-associated clonal expansion in the hematopoietic system is not clear. However, the prevalence of *PPM1D* somatic mutations in other tissues such as the esophagus, suggests the need for further investigation (Yokoyama et al., 2019 [DOI](#)).

Materials and Methods

Cell lines and Reagents

Cas9-expressing OCI-AML2 cells were generated by lentiviral transduction using pKLV2-EF1aBsd2ACas9-W plasmid obtained from Dr. Kosuke Yusa from the Sanger Institute (Addgene #67978). Four days post-transduction, cells underwent blasticidin selection. Single clones were obtained by fluorescence-activated cell sorting and functionally tested for Cas9 activity using a lentiviral reporter pKLV2-U6gRNA5(gGFP)-PGKBFP2AGFP-W (Addgene #67980). *PPM1D*-mutant cell lines were generated using the RNP-based CRISPR/Cas9 delivery method using a single sgRNA (GCTAAAGCCCTGACTTTA). Single cells were sorted into 96-well, round-bottom plates and expanded. Clones were validated by Sanger sequencing, TIDE analysis, and western blot to visualize the overexpressed, truncated mutant protein. Two validated *PPM1D*-mutant clones were selected for the CRISPR dropout screen.

CRISPR Dropout Screen and Analyses

For large-scale production of lentivirus, 15 cm plates of 80-90% confluent 293T cells were transfected using Lipofectamine 2000 (Invitrogen) with 7.5 ug of the Human Improved Whole-Genome Knockout CRISPR library V1 (by Kosuke Yusa, Addgene #67989), 18.5 ug of psPax2, and 4 ug of pMD2.G. A lentivirus titer curve was performed prior to the screen to determine the volume of viral supernatant to add for a multiplicity of infection (MOI) of ~0.3. For the CRISPR dropout screen, one WT and two independent *PPM1D* mutant Cas9-expressing OCI-AML2 cell lines were used as biological replicates, with three technical replicates per line. 3×10^7 cells were transduced with the lentivirus library supernatant. Three days post-transduction, the cells were selected with puromycin for three days. Cells were collected on day 28 for genomic DNA isolation using isopropanol precipitation. Illumina adapters and barcodes were added to samples by PCR as previously described (Tzelepis et al., 2016 [DOI](#)). Single-end sequencing was performed on the HiSeq 2000 V4 platform and cell-essential genes were identified using the MaGECK-VISPR (Li et al., 2014 [DOI](#)).

Competitive Proliferation Assay

Gene-specific sgRNAs were cloned into the pKLV2-U6gRNA5(BbsI)-PGKpuro2ABFP (Addgene #67974) lentiviral backbone. 293T cells (0.4×10^6 cells per well) were seeded in a six-well plate the day prior and transfected using lipofectamine 3000 with pMD2G (0.8 ug), pAX2 (1.6 ug), and the sgRNA-BFP (1.6 ug) plasmids. Cas9-expressing cells were then seeded in 12-well plates (200k cells per well, in triplicates) in media supplemented with 8ug/ml polybrene and 5 ug/mL blasticidin, and lentivirally transduced at a titer that yields 50% infection efficiency. Cells were assayed using flow cytometry for BFP expression between 4- and 16-days post-transduction and normalized to the BFP percentage at day 4.

Drug and Proliferation Assays

Drug and proliferation assays were done using the Cell Proliferation MTT Kit (Sigma) as per manufacturer's protocol. Briefly, 1×10^4 cells were plated in 96-well, flat bottom plates and treated with vehicle or drugs in a total volume of 100 μ L. Plates were incubated at 37°C for at least 24 hours. 10 μ L of MTT labeling reagent was added to each well and incubated for 4 hours. 100 μ L of solubilization buffer was added to each well and incubated overnight. Plates were analyzed using a fluorometric microplate reader at 550 nm. Stock solutions of ATN-224 (Cayman Chemical #23553) and LCS-1 (MedChem, HY-115445) were in DMSO and frozen in -20°C .

SOD Activity Assay

SOD activity was measured per manufacturer's protocol (Invitrogen, Cat#EIASODC). Briefly, cells were treated with low or high-dose ATN-224 (0.625 μM and 1.25 μM , respectively) for 16-hours and harvested. Cells were washed with PBS and lysed with ice cold NP-40 lysis buffer (Invitrogen #FNN0021) with protease inhibitor (ThermoFisher, #78440). Cells were sonicated for 5 seconds $\times$ 5 rounds and then spun at 13,000 rpm for 10 minutes at 4°C. Protein concentrations were measured using BCA assay (ThermoFisher, #23225) and diluted to a concentration of 10 $\mu\text{g}/\mu\text{L}$. 100 μg (10 μL) of protein was loaded per sample and incubated for 20 minutes with the added substrates. Plates were read on a microplate reader at 450 nm.

Intravenous transplantation of leukemia cells in NSG mice

WT and PPM1D-mutant OCI-AML2 cells were transduced with empty vector or sgSOD1 lentivirus, as described in the "Competitive Proliferation Assay" methods section above. Three days post-transduction, cells underwent puromycin selection (3 $\mu\text{g}/\text{mL}$). On day six post-transduction, the infection rate was determined by flow cytometry using the percentage of BFP⁺ cells. All samples had an infection rate of >95%. 8-week-old male NOD.Cg-Prkdc^{scid}Il2rg^{tm1Wjl}/SzJ (NSG) mice were purchased from The Jackson Laboratory (strain #005557) and sublethally irradiated (250 cGy) immediately prior to transplantation. 2×10^6 cells were intravenously injected in the tail vein of mice (n=8 per group). After transplantation, mice were monitored daily for disease progression and humane euthanasia was performed when animals lost >15% body weight or had signs of severe disease (limb paralysis, decreased activity, and hunching). All animal procedures and studies were done in accordance with the Institutional Animal Care and Use Committee (IACUC).

Alkaline Comet Assay

Comet assays were conducted as previously described (Greve et al., 2012 [DOI](#); Schmezer et al., 2001 [DOI](#)). Cells were resuspended to 1×10^5 cells/mL and mixed with 1% low-melting agarose (R&D Systems) at a 1:10 ratio and plated on 2-well comet slides (R&D Systems). Cells were then lysed overnight and immersed in alkaline unwinding solution as per manufacturer's protocol (Trevigen). Fluorescence microscopy was performed at 10X magnification using the Keyence BZ-X800 microscope and analyses of comet tails were performed using the Comet Assay IV software (Instem). At least 150 comet tails were measured per sample.

Chromosome aberration analysis mitotic chromosome spreads

Primary resting mouse splenic B-cells were isolated using anti-CD43 microbeads (Miltenyi Biotec) and activated with 25 $\mu\text{g}/\text{mL}$ LPS (Sigma), 5 ng/mL IL-4 (Sigma), and 0.5 $\mu\text{g}/\text{mL}$ anti-CD180 (BD Pharmingen) for 30 hours. The cells were then treated with cisplatin for 16 hours at two concentrations - 0.5 μM and 5 μM cisplatin. Metaphases were prepared as previously described (Zong et al., 2019 [DOI](#)). Briefly, cells were arrested at mitosis with colcemid (0.1 $\mu\text{g}/\text{mL}$, ThermoFisher) for 1 hour. Cells were then incubated in a prewarmed, hypotonic solution of potassium chloride (75 mM) for 20 minutes to induce swelling and fixed in methanol/glacial acetic acid (3:1). Droplets were spread onto glass slides inside a cytogenetic drying chamber.

Fluorescence *in situ* hybridization was performed using a Cy3-labeled peptide nucleic acid probe to stain telomeres and DNA was counterstained by DAPI. At least 50 metaphases were scored for chromosome aberrations for each experimental group.

ROS Assays

To measure superoxide, total cellular reactive oxygen species (ROS), and lipid peroxidation, 1×10^6 cells were collected after the indicated treatments and washed with PBS. The cells were stained with 1 μ M Mitosox Green (Thermofisher), 5 μ M dihydroethidium (Thermofisher), 20 μ M 2'-dichlorofluorescein diacetate (DCFDA, Abcam), or 2.5 μ M BODIPY 581/591 (Thermofisher) in FBS-free Hanks' buffered saline solution (HBSS, Thermofisher), and incubated at 37°C for 30 minutes. The staining was quenched with flow buffer (PBS, 2% FBS, 1% HEPES) and washed twice before resuspension in DAPI-containing flow buffer to assess ROS in viable cells. For detection of intracellular glutathione (GSH), we utilized the Intracellular glutathione detection assay kit (abcam) as per manufacturer's protocol. The data was acquired using an LSRII (BD Biosciences) and analyzed on Flowjo. The mean fluorescence intensity (MFI) was used for data analysis.

Reverse-Phase Protein Array

Reverse phase protein array assays for antibodies to proteins or phosphorylated proteins in different functional pathways were carried out as described previously (Coarfa et al., 2021 [DOI](#); Lu H.-Y., 2021 [DOI](#); Wang et al., 2022 [DOI](#)). Specifically, protein lysates were prepared from cultured cells with modified Tissue Protein Extraction Reagent (TPER) (Life Technologies Corporation, Carlsbad, CA) and a cocktail of protease and phosphatase inhibitors (Roche, Pleasanton, CA) (Lu et. al. 2021). The lysates were diluted into 0.5 mg/ml in SDS sample buffer and denatured on the same day. The Quanterix 2470 Arrayer (Quanterix, Billerica, MA) with a 40 pin (185 μ m) configuration was used to spot samples and control lysates onto nitrocellulose-coated slides (Grace Bio-labs, Bend, OR) using an array format of 960 lysates/slide (2880 spots/slide). The slides were processed as described and probed with a set of 264 antibodies against total proteins and phosphoproteins using an automated slide stainer Autolink 48 (Dako, Santa Clara, CA). Each slide was incubated with one specific primary antibody and a negative control slide was incubated with antibody diluent without any primary antibody. Primary antibody binding was detected using a biotinylated secondary antibody followed by streptavidin-conjugated IRDye680 fluorophore (LI-COR Biosciences, Lincoln, NE). Total protein content of each spotted lysate was assessed by fluorescent staining with Sypro Ruby Protein Blot Stain according to the manufacturer's instructions (Molecular Probes, Eugene, OR).

Fluorescence-labeled slides were scanned on a GenePix 4400 AL scanner, along with accompanying negative control slides, at an appropriate PMT to obtain optimal signal for this specific set of samples. The images were analyzed with GenePix Pro 7.0 (Molecular Devices, Silicon Valley, CA). Total fluorescence signal intensities of each spot were obtained after subtraction of the local background signal for each slide and were then normalized for variation in total protein, background and non-specific labeling using a group-based normalization method as described (Lu H.-Y., 2021 [DOI](#)). For each spot on the array, the-background-subtracted foreground signal intensity was subtracted by the corresponding signal intensity of the negative control slide (omission of primary antibody) and then normalized to the corresponding signal intensity of total protein for that spot. Each image, along with its normalized data, was evaluated for quality through manual inspection and control samples. Antibody slides that failed the quality inspection were either repeated at the end of the staining runs or removed before data reporting. A total of 261 antibodies remained in the list. Multiple t-tests with Benajimini Hochberg correction were performed for statistical analysis and filtering was based on an FDR <0.2 and linear fold change of >1.25.

RNA-sequencing

Bulk RNA-sequencing was performed on WT and *PPM1D*-mutant OCI-AML2 cells after lentiviral *SOD1* CRISPR knockout. Cells were transduced with pKLV2-U6-sgRNA-BFP lentivirus (either empty vector or with *SOD1*-sgRNA). Transduced cells were then cultured for 10 days and BFP+ cells were sorted directly into Buffer RLT Plus with β -mercaptoethanol. RNA was isolated using the Allprep DNA/RNA Micro Kit (Qiagen) per manufacturer's protocols. RNA-sequencing library preparation was done using the True-Seq Stranded mRNA kit (Illumina) per manufacturer's protocol. Quality control of libraries was performed using a TapeStation D1000 ScreenTape (Agilent, 5067-5584). Libraries were then sequenced using an Illumina Nextseq 2000 sequencer, aiming for >20 million reads per biological replicate. Paired-end RNA-sequencing reads were obtained, obtained and trimmed using trimGalore (<https://github.com/FelixKrueger/TrimGalore>). Mapping was performed using the STAR package (Dobin et al., 2013) against the human genome build UCSC hg38 and counts were quantified with featureCounts (Liao et al., 2014). Differential expression analysis was performed using the DESeq2 R package (1.28.1) (Love et al., 2014). P-values were adjusted with Benjamini and Hochberg's approach for controlling the false discovery rate (FDR). Significant differentially expressed genes between the indicated comparisons were filtered based on an FDR<0.05 and absolute fold change exceeding 1.5. Pathway enrichment analysis was carried out using the GSEA (<http://software.broadinstitute.org/gsea/index.jsp>) software package and significance was achieved for adjust FDR<0.25.

Seahorse Assay

Mitochondrial bioenergetics in AML cell lines were performed using the Seahorse XFp Cell Mito Stress Kit (Agilent Technologies) on the Seahorse XFe96 Analyzer. Cells were resuspended in XF RPMI base media supplemented with 1 mM pyruvate, 2 mM L-glutamine, 10 mM glucose. 1×10^5 cells/well were seeded in poly-D-lysine (ThermoFisher) coated XFe96 plates. The plate was incubated in a non-CO₂ incubator at 37°C for 1 hour to equilibrate. OCR and ECAR measurements were taken at baseline and every 8 minutes after sequential addition of oligomycin (2 μ M), FCCP (0.5 μ M), and rotenone/ antimycin A (0.75 μ M). All measurements were normalized to the number of viable cells.

Generation of *PPM1D* WT U2OS cells using CRISPR editing

U2OS cells containing the DR-GFP (for homologous recombination) or EJ5-GFP (for non-homologous end-joining) DNA repair reporter cassettes were kindly provided by the Bertuch Lab at Baylor College of Medicine. To establish *PPM1D*-WT isogenic lines, knock-in CRISPR editing was performed with a single-stranded oligodeoxynucleotide (ssODN) template: TGCCCTGGTTC GTAGCAATGCCTTCTCAGAGAATTTCTAGAGGTTTCAGCTGAGATAGCTCGTGAGAATGT ACAAGGTGTAGTCATACCCTAAAAGATCCAGAACCACTGAAGAAAATGCGCTAAAGCCCT GACTTTAAGGATACA. The *PPM1D* sgRNA sequence used was: ATAGCTCGAGA GAATGTCCA. 1.3 μ g of Cas9 (IDT) was incubated with 1 μ g of sgRNA for 15 minutes at room temperature. 1 μ g of the ssODN template was then added to the Cas9-sgRNA complexes and mixed with 20,000 U2OS cells and resuspended in 10 μ L of Buffer R, immediately prior to electroporation. The neon electroporation system was used with the following conditions: 1400v, 15 ms, 4 pulses. Single cell-derived clones were genotyped by Sanger sequencing and *PPM1D* protein expression was validated by western blot.

GFP reporter-based DNA repair assays

For the DNA repair reporter assay, 100,000 U2OS cells were seeded in a 12-well plate in antibiotic-free Dulbecco's Modified Eagle Medium (DMEM, ThermoFisher) supplemented with 10% FBS. Cells were transfected with 3.6 μ L of Lipofectamine 2000 (Invitrogen) in 200 μ L of OptiMEM with 0.8 μ g

of the I-SceI expression plasmid (pCBASce, Addgene #60960). The media was replaced the next morning and the cells were trypsinized 48-hours post-transfection for analysis of GFP expression by flow cytometry (BD Biosciences).

Immunofluorescence microscopy

12 mm glass coverslips were coated with 50 $\mu\text{g/mL}$ poly-D-lysine (ThermoFisher) for 30 minutes at room temperature and washed with sterile PBS. 0.5×10^6 suspension cells/well were seeded on coverslips and incubated for 1 hour at 37°C to allow for adherence. Samples were then fixed with 4% paraformaldehyde for 10 minutes at 37°C and washed three times with 0.01% Triton-X PBS (PBS-T). Fixed cells were permeabilized with 0.5% PBS-T for 20 minutes, washed three times, and incubated with 5% goat serum (ThermoFisher) for 1 hour at room temperature. Afterwards, samples were incubated overnight at 4°C with the following primary antibodies: rabbit anti-Rad51 (Cell Signaling #8875S 1:100) or rabbit anti-53BP1 (ThermoFisher #PA1-16565, 1:500). The following day, samples were washed and incubated at room temperature for 1 hour with Alexafluor 488-conjugated goat anti-rabbit IgG (#111-545-144, Jackson ImmunoResearch, 1:500). After secondary antibody incubation, the coverslips were washed three times with PBS and mounted with fluoromount-G mounting medium with DAPI (ThermoFisher) on glass microscope slides and sealed with nail polish. Imaging was done on the Keyence BZ-X800 microscope and foci analysis was performed using CellProfiler.

Immunoblotting

Cells were lysed with 1x RIPA buffer supplemented with Halt Protease and Phosphatase inhibitor cocktail (ThermoFisher) for 1 hour at 4°C . Protein concentration was quantified using the Pierce BCA protein assay kit (ThermoFisher) and boiled at 95°C in 1x Laemmli (Biorad) for 7 minutes. The samples in which mitochondrial proteins were probed were not boiled, as boiling can cause signal reduction. Instead, samples were warmed to 37°C for 30 minutes prior to loading. The proteins were separated by SDS-PAGE on 4-15% gradient gels (Biorad) and transferred on to PVDF membranes using the iBlot Dry Blotting system (ThermoFisher). Membranes were incubated for 1 hour at room temperature in 5% milk in Tris-buffered saline solution with Tween-20 (TBST). After washing, the membranes were incubated overnight at 4°C with the following primary antibodies: mouse anti-PPM1D (F-10, Santa Cruz, 1:1000), mouse anti-GAPDH (MAB374, Millipore, 1:200), mouse total OXPHOS Human antibody cocktail (ab110411, Abcam, 1:1000), mouse anti-Vinculin (V9131, Sigma Aldrich, 1:2000). The following day, membranes were washed twice with TBST and incubated for 1 hour with HRP-linked anti-rabbit IgG or anti-mouse IgG (Cell Signaling, 1:5000 – 1:10,000) at room temperature. Blots were imaged on the Bio-Rad ChemiDoc platform.

Statistical analysis

Statistical analysis incorporated in the MaGECK-VISPR algorithm includes p-value and FDR calculations. GraphPad Prism 6.0 was used for other statistical analyses. The sample size (n) specified in the Figure Legends was used for statistical analysis and denotes the number of independent biological replicates. The main conclusions were supported by data obtained from at least two biological replicates. The graphs presented in the figures are shown with error bars indicating either mean $\pm$ SEM or mean $\pm$ SD, as mentioned in the Figure Legends. Two-tailed t-tests were performed to calculate statistics, assuming unequal standard deviations, unless mentioned otherwise. Significance levels are indicated in the figures and were determined using GraphPad PRISM. Results were considered statistically significant at * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

Acknowledgements

This work was supported by R01CA237291 and P01CA265748. This work was also supported by the NCI Cancer Center Support Grant P30CA125123 which partly supports the Cytometry Core, the Proteomics & Metabolomics Core, and the Antibody-based Proteomics Core. Support for the cores was also provided by the Cancer Prevention and Research Institute of Texas (CPRIT) from grants: RP180672, RR024574, and RP210227 and NIH S10OD028648. LZ was supported by the Baylor Research Advocates for Student Scientists (BRASS) Foundation, and the Janice McNair Medical Foundation.

Author Contributions

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Investigation: LZ, JIH, EDB, CWC, AGG, AGM, SMW, EC, RM, RR, AN, SJ, SH, JNA

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Computational analysis: EDB, CWC, TDP

Supervision: JIH, AN, BBadV, CC, JNA, KT, GV, MAG

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Contributed unpublished, essential data, or reagents: KT, HU, AGG, AGM, RM, RR, JNA

Competing Interests

The authors declare they have no competing interests.

Data and materials availability

All data are available in the main text or in the supplementary materials. All raw and processed data generated in this work will be publicly available at GEO data repositior pending scientific review. The human cell lines generated by the Goodell laboratory for this study are available upon request and will require a standard Materials Transfer Agreement (MTA). Any additional information required to analyze the data reported in this paper is available from the lead contact upon request.

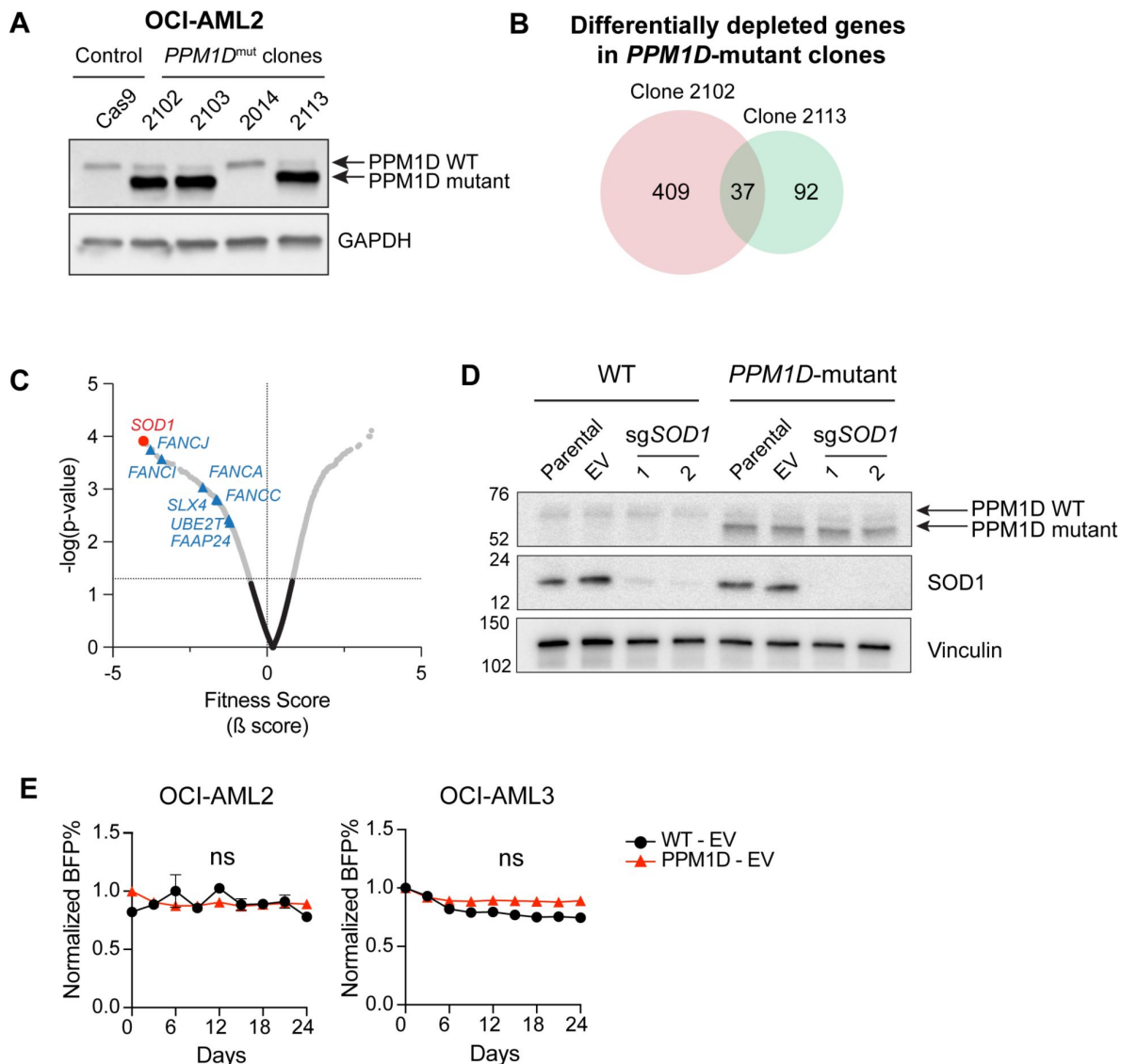


Figure 1 – Figure Supplement 1.

***SOD1* is a synthetic lethal vulnerability of *PPM1D*-mutant leukemia cells.**

(A) Immunoblot validation of *PPM1D*-mutant Cas9-expressing OCI-AML2 cells generated and used for CRISPR screening. Blots were probed with anti-PPM1D (1:1000) and GAPDH (1:1000). Clones 2102 and 2113 were selected for the dropout screen. **(B)** Venn diagram of genes that were depleted from the two *PPM1D*-mutant clones (#2102, 2113) used in the dropout screen, but not depleted in the WT control lines. 37 genes were found to be depleted in both mutant clones. For a full list of genes, see Figure 1–source data 1. **(C)** Volcano plot of synthetic lethal hits ranked by fitness score with the Fanconi Anemia pathway genes highlighted in blue. **(D)** Immunoblot validation of *SOD1*-deletion. WT and *PPM1D*-mutant Cas9-OCI-AML2 cells were transduced with control (EV) or sg*SOD1* lentiviruses. Two sgRNAs targeting *SOD1* were tested. Three days post-transduction, the cells underwent puromycin selection (3 μ g/mL) for three days after which they were harvested for western blot. Blots were probed with anti-PPM1D (1:1000), anti-SOD1 (1:500), and anti-vinculin (1:2500). **(E)** Cas9-OCI-AML2 and Cas9-OCI-AML3 WT or *PPM1D*-mutant cells were transduced with the empty vector control backbone tagged with a blue fluorescent protein (BFP) reporter. Cells were assayed by flow cytometry between 3- and 24-days post-transduction and normalized to the BFP percentage at day 3. Data shown are mean $\pm$ SD (n=2 per condition).

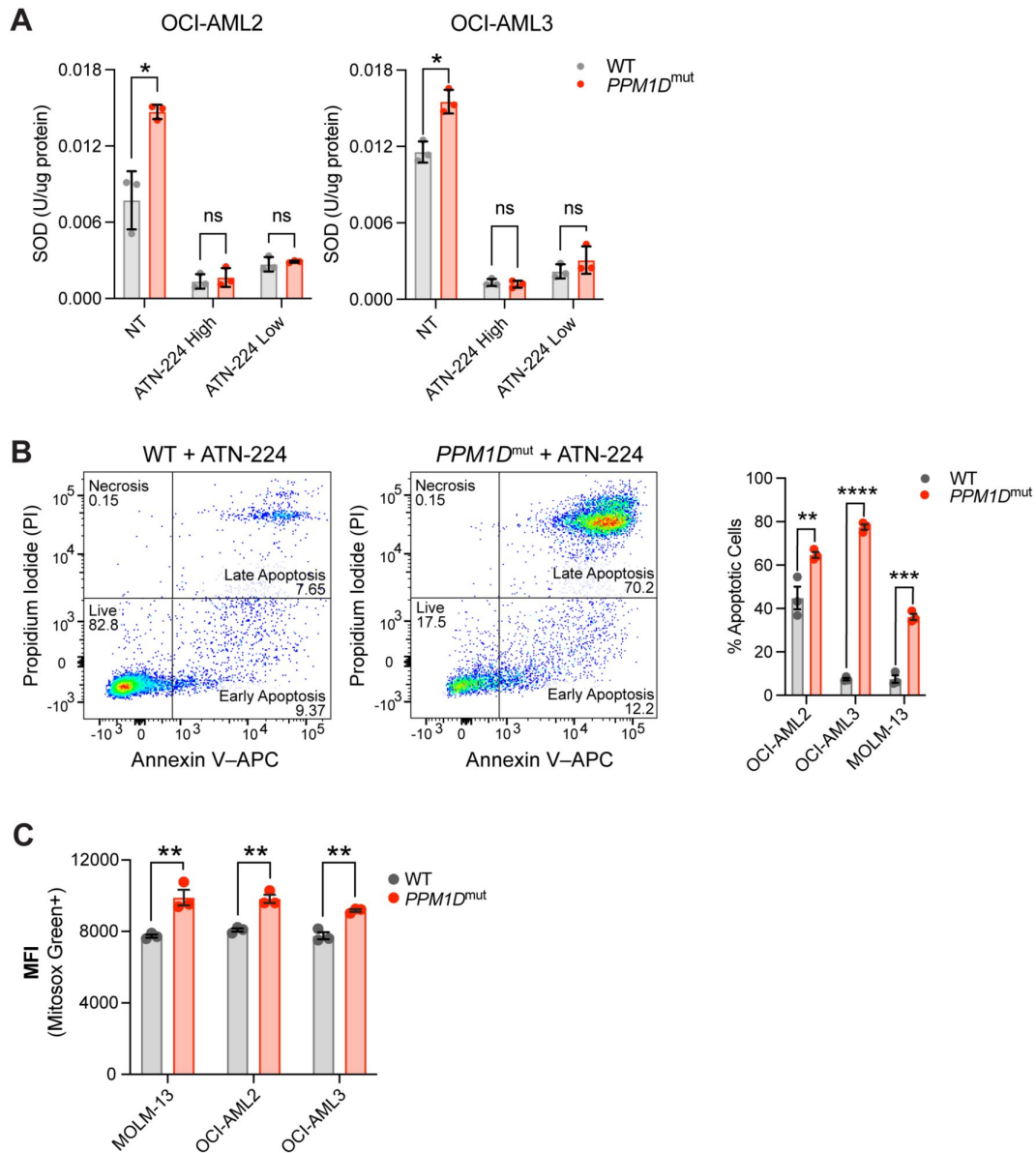


Figure 2 – Figure Supplement 1.

***PPM1D*-mutant cells have increased oxidative stress.**

(A) SOD activity assays in OCI-AML2 and OCI-AML3 cells at baseline (NT), or treated with high (12.5 μ M) or low (6.25 μ M) doses of ATN-224 for 16 hours. **(B)** Left: Representative flow cytometry plots of WT and *PPM1D*-mutant cells treated with ATN-224 (25 μ M for 24 hours) and stained for Annexin V-APC and PI for apoptosis; multiple unpaired t-tests, ns=non-significant, * p <0.05, ** p <0.01, *** p <0.001, **** p <0.0001. **(C)** Endogenous mitochondrial superoxide levels of WT and *PPM1D*-mutant leukemia cell lines were measured using MitoSox Green staining (1 μ M). The mean fluorescence intensity (MFI) of MitoSox Green was measured by flow cytometry. Mean $\pm$ SD (n =3) is shown.

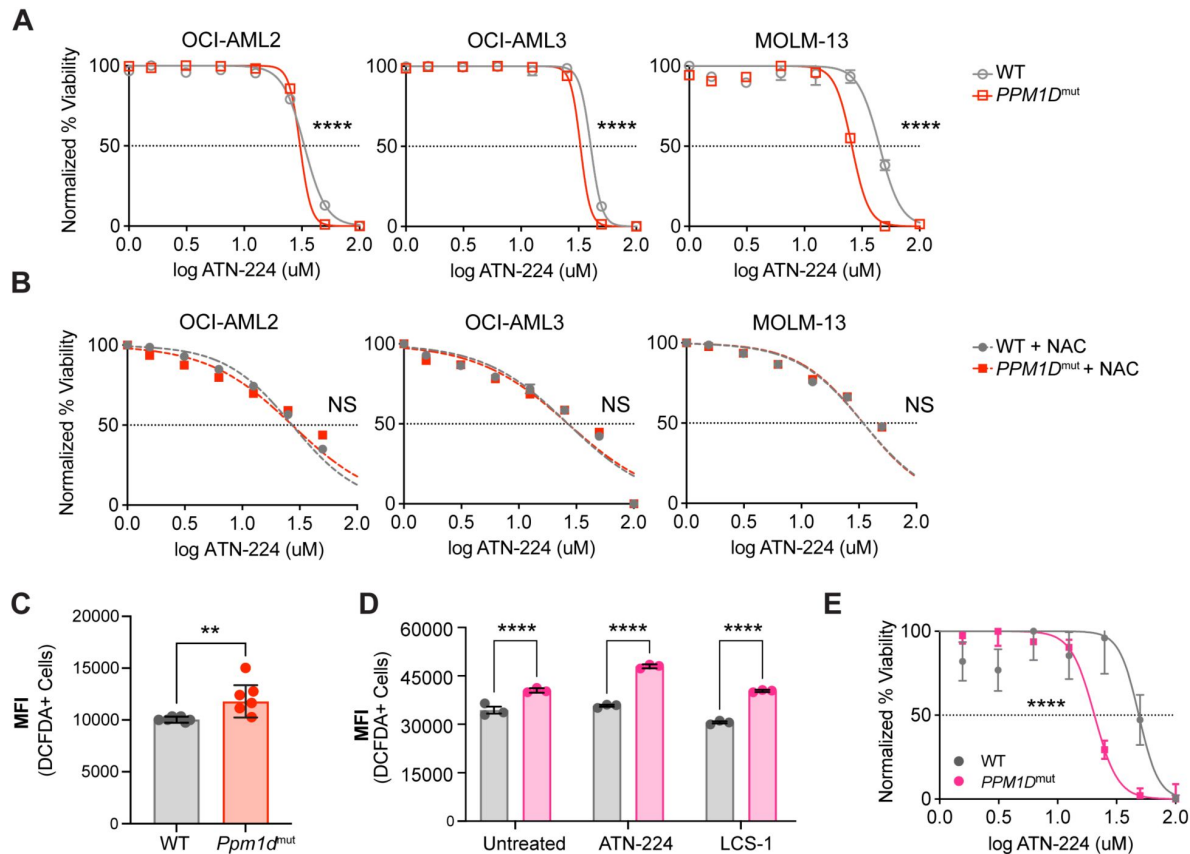


Figure 2 – Figure Supplement 2.

PPM1D-mutant cells have increased oxidative stress.

(A–B) Dose response curves for cell viability with SOD1-inhibitor (ATN-224) **(A)** or ATN-224 in combination with 0.25 μ M NAC **(B)** in WT and *PPM1D*-mutant leukemia cell lines after 24-hours. Mean $\pm$ SD ($n=3$) is shown along with a non-linear regression curve. All values are normalized to the baseline cell viability with vehicle, as measured by MTT assay. **(C)** Total reactive oxygen species (ROS) of WT and *Ppm1d*-mutant MEFs measured by DCFDA (10 μ M) staining. Mean fluorescence intensity (MFI) was determined by flow cytometry. $n=6$ biological replicates were used for each genotype. Data shown are the mean of each biological replicate; unpaired t-test. **(D)** Total ROS of WT GM12878 (grey) and *PPM1D*-mutant (pink) patient lymphoblastic cell lines (LCLs) at baseline, and after 24-hrs of SOD1 inhibition measured by DCFDA (10 μ M) staining. MFI was determined by flow cytometry; multiple unpaired t-tests. **(E)** Dose response curve of WT and *PPM1D*-mutant LCLs after ATN-224 treatment. IC50s of WT and *PPM1D*-mutant LCLs were 48.8 μ M and 20.51 μ M, respectively as measured by MTT assay; non-linear regression analysis, ns=non-significant ($p>0.05$), ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$.

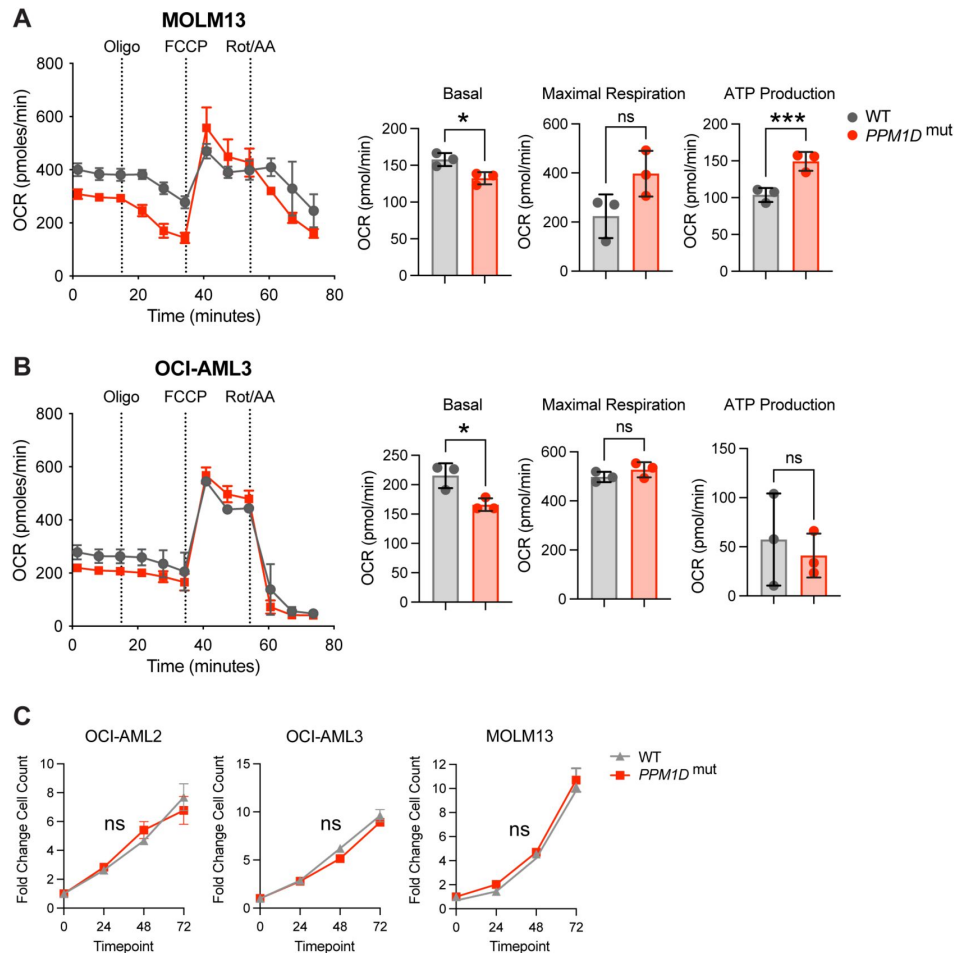


Figure 3 – Figure Supplement 1.

***PPM1D*-mutant cells have altered mitochondrial function.**

(A,B) Measurement of mitochondrial oxygen consumption rate (OCR) by seahorse assay in WT vs. *PPM1D*-mutant MOLM-13 (A) and OCI-AML3 (B) cells after treatment with oligomycin (1.5 μ M), FCCP (0.5 μ M), and rot/AA (0.5 μ M). Quantification of basal, maximal, and ATP-linked respiration shown. Each cell line was performed in technical triplicates, student's t-test. (C) Growth curves of WT and *PPM1D*-mutant leukemia cell lines at 24-, 48-, and 72-hours. Cell counts were normalized to day 0. ns=non-significant ($p > 0.05$), * $p < 0.05$, *** $p < 0.001$.

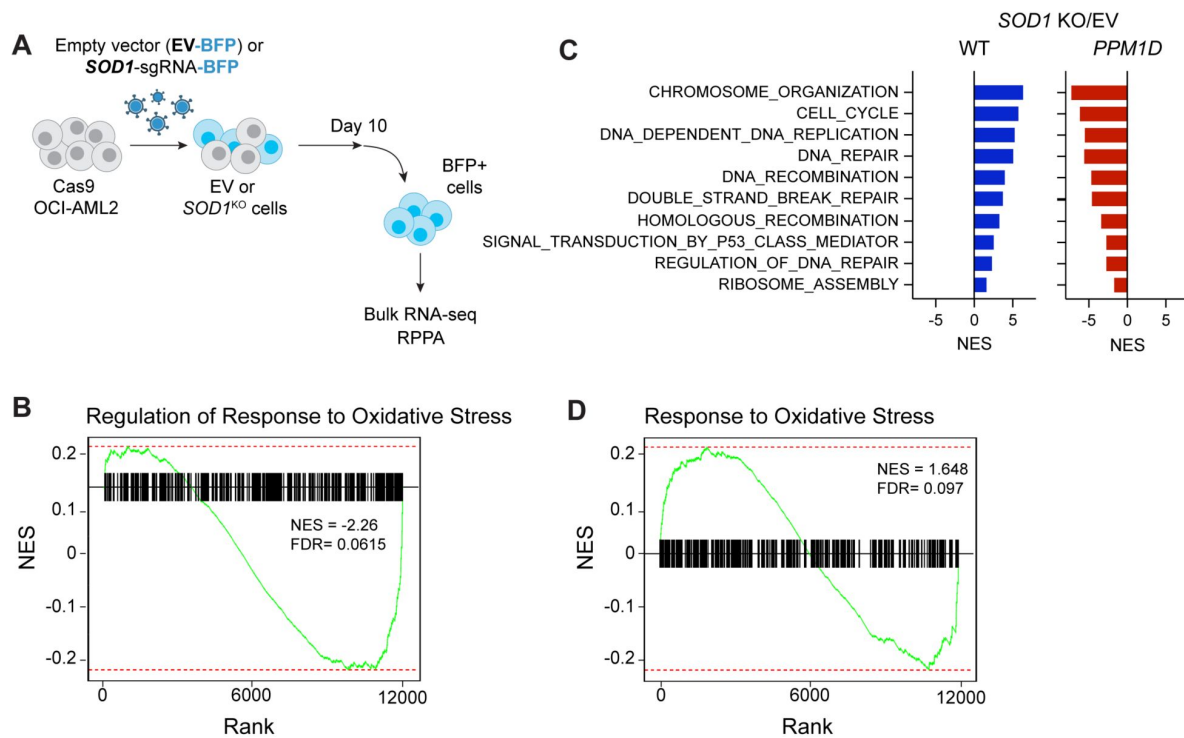


Figure 4 – Figure Supplement 1.

***PPM1D*-mutant cells have reduced oxidative stress response.**

(A) Schematic of the experimental setup for the bulk RNA-sequencing and reverse-phase protein array. WT and *PPM1D*-mutant Cas9 OCI-AML2 cells were transduced with either empty vector (EV)-BFP or *SOD1*-sgRNA-BFP. Cells were passaged for ten days and then sorted for BFP expression for downstream analysis. (B, D) GSEA enrichment plots for *PPM1D*-mutant cells compared to WT after transduction with EV (B) or after *SOD1*-knockout (D) for the “Regulation of Response to Oxidative Stress” (GO:1902882) and “Response to Oxidative Stress” (GO:0006979). NES are shown with FDR<0.25. (C) GSEA analysis of RNA-sequencing of *SOD1*-deleted cells compared to EV control in WT and *PPM1D*-mutant cells. Blue and red bars indicate significantly up- and downregulated pathways, respectively. Normalized enrichment scores (NES) are indicated. All pathways filtered for FDR<0.25.

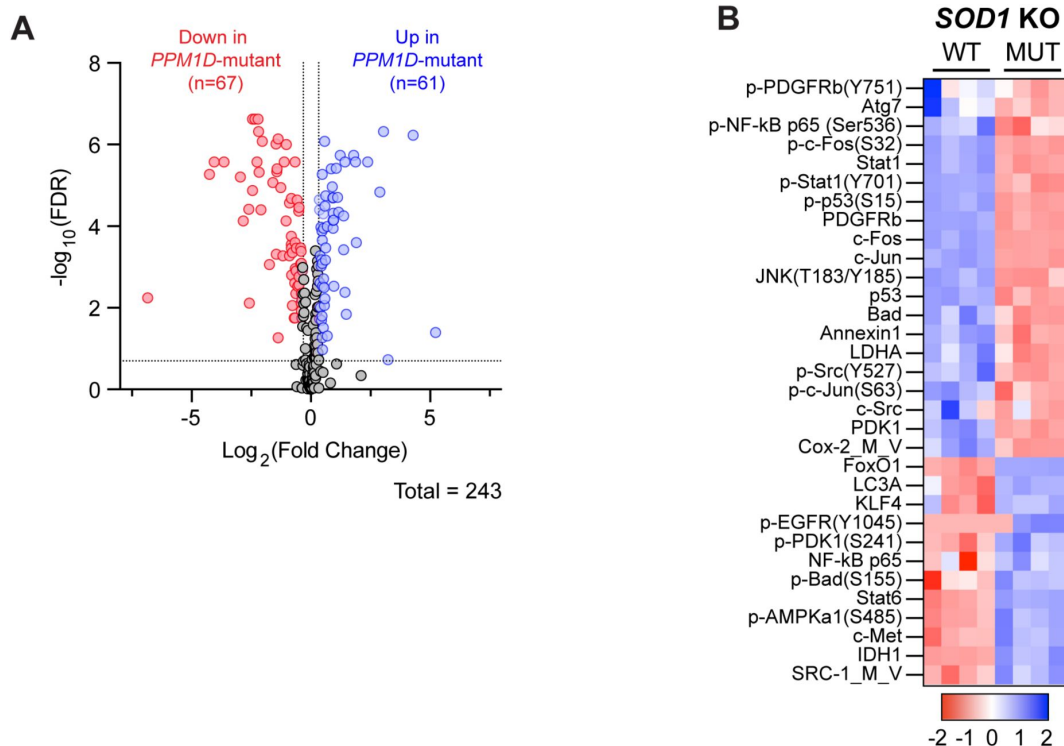
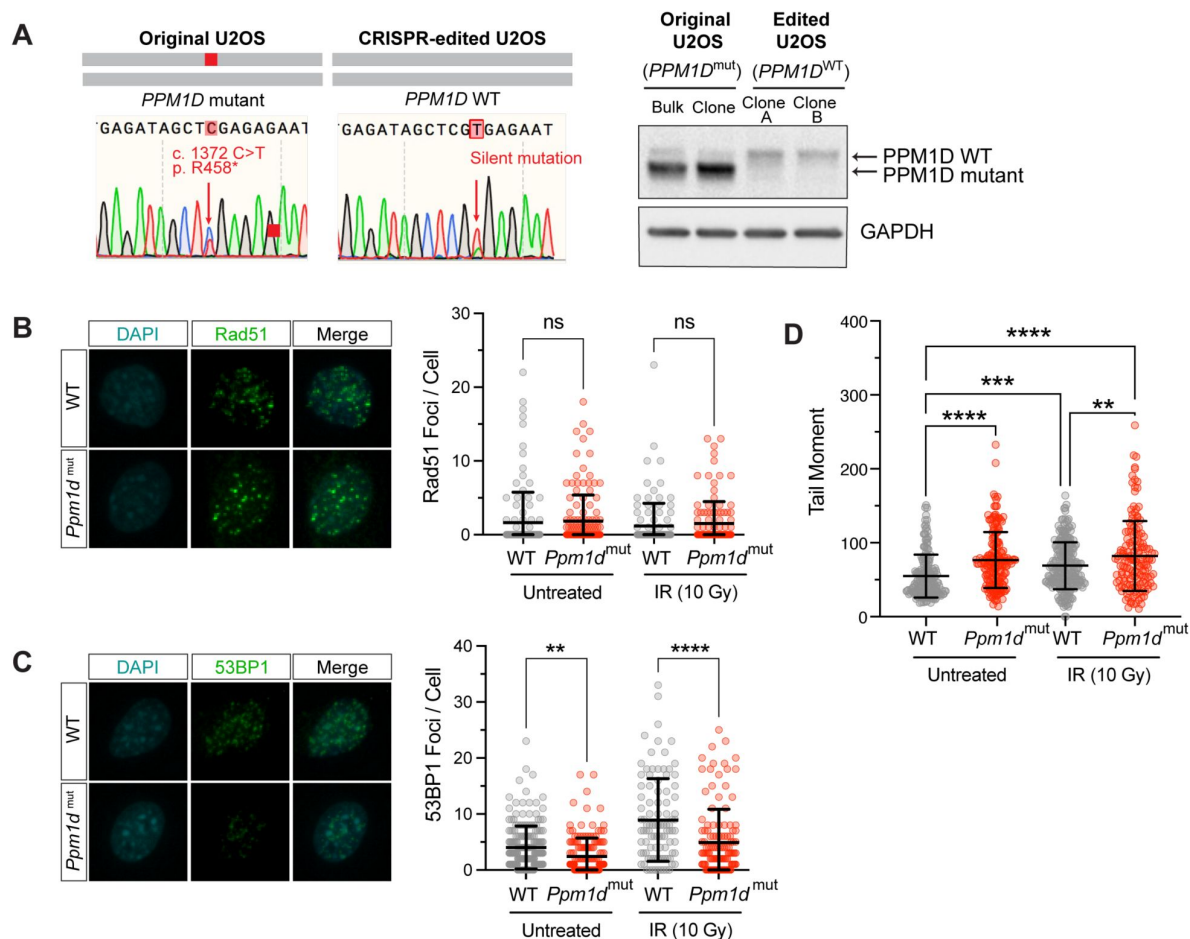


Figure 4 – Figure Supplement 2.

***PPM1D*-mutant cells have reduced oxidative stress response.**

(A) Volcano plot of the differentially expressed proteins from the RPPA in *PPM1D*-mutant OCI-AML2 cells compared to WT. Red and blue dots indicate significantly up- or downregulated proteins, respectively, with a cutoff FDR<0.2 and linear fold change >|1.2|. **(B)** RPPA profiling of WT and *PPM1D*-mutant cells after *SOD1* deletion. Proteins from the “Response to Oxidative Stress” pathway have been selected for the heatmap. Each column represents a technical replicate. See Figure 4-source data 2 for the raw data.



Supplementary Figure 5.

PPM1D-mutations increase genomic instability and impairs non-homologous end-joining repair.

(A) Left: Sanger sequencing traces of the parental U2OS cell line harboring a c.1372 C>T mutation in *PPM1D* and the CRISPR-edited U2OS cell line with mutation corrected to WT *PPM1D*. Right: Immunoblot validation of these clones are shown. Lysates were probe with anti-PPM1D (1:1000) and anti-GAPDH (1:1000). (B,C) Left: Representative images of Rad51 and 53BP1 immunofluorescence microscopy. Mouse embryonic fibroblasts were treated with 10 Gy irradiation, harvested 1-hour post-irradiation and stained for the indicated markers. Right: Quantification of the number of foci per cell is shown. Analysis was performed using CellProfiler. n>100 cells for each condition; students t-test. (D) Comet assay quantification of mouse embryonic fibroblasts at baseline and after 1-hour post-irradiation (10 Gy). Quantification and analyses of tail moments were performed using the Comet IV software. n≥150 comets were scored per experimental group; 2way ANOVA, ns=non-significant (p>0.05), *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

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Reviewer #1 (Public Review):

Summary:

Gain-of-function mutations and amplifications of PPM1D are found across several human
cancers and are associated with advanced tumor stage, worse prognosis, and increased
lymph node metastasis. This manuscript presents important findings that SOD1 inhibition is a
potential strategy to achieve therapeutic synergism for PPM1D-mutant leukemia; and
demonstrates the redox landscape of PPM1D-mutant cells.

Strengths:

In this manuscript, Zhang and colleagues investigate the synthetic-lethal dependencies of PPM1D (protein phosphatase, Mg²⁺/Mn²⁺ dependent 1D) in leukemia cells using CRISPR/Cas9 screening. They identified that SOD1 (superoxide dismutase-1) as the top hit, whose loss reduces cellular growth in PPM1D-mutant cells, but not wildtype (WT) cells. Consistently, the authors demonstrate that PPM1D-mutant cells are more sensitive to SOD1 inhibitor treatment. By performing different in vitro studies, they show that PPM1D-mutant leukemia cells have elevated level of reactive oxygen species (ROS), decreased basal respiration, increased genomic instability, and impaired non-homologous end-joining repair. These data highlight the potential of SOD1 inhibition as a strategy to achieve therapeutic synergism for PPM1D-mutant leukemia; and demonstrates the redox landscape of PPM1D-mutant cells.

Weaknesses:

While the current study has identified synthetic lethality of PPM1D-mutant leukemia cells upon SOD1 inhibition, the underlying mechanism remains elusive. Although ROS levels have been assessed between wild-type (WT) and PPM1D-mutant leukemia cells, the specific redox alterations induced by SOD1 inhibition in PPM1D mutant versus WT cells have not been elucidated. To address this gap, direct comparisons of ROS levels using various probes should be conducted between PPM1D mutant and WT cells under conditions of SOD1 inhibition.

<https://doi.org/10.7554/eLife.91611.2.sa2>

Reviewer #2 (Public Review):

The authors used a whole genome CRISPR screen to identify targetable synthetic lethalties associated with PPM1D mutations, known poor prognosis and currently undruggable factors in leukemia. The authors identified the cytosolic superoxide dismutase (SOD1, Cu/Zn SOD) as a major protective factor in PPM1D mutant vs. wt cells, and their study investigates associated mechanisms of this protection. Using both genetic depletion and small molecule inhibitors of SOD1, the authors conclude that SOD1 loss exacerbates mitochondrial dysfunction, ROS levels and DNA damage phenotypes in PPM1D mutant cells, decreasing cell growth in AML cells. The data strongly support that PPM1D mutant cells have high levels of total peroxides and elevated DNA breaks, and that genetic depletion of SOD1 decreases cell growth in two AML cell lines. However, the authors don't explain how superoxide radical (which is not damaging by itself) induces such damage, the on-target effects of the SOD1 inhibitors at the concentrations is not clear, the increase in total hydroperoxides is not supported by loss of SOD1, the changes in mitochondrial function are small, and there is no assessment of how the mitochondrial SOD2 expression or function, which dismutates mitochondrial superoxide, is altered. Overall these studies do not distinguish between signal vs. damaging aspects of ROS in their models and do not rule out an alternate hypothesis that loss of SOD1 increases superoxide production by cytosolic NADPH activity which would significantly alter ROS-driven regulation of kinase/phosphatase signal modulation, affecting cell growth and proliferation as well as DNA repair. Additionally, with the exception of growth defects demonstrated with sgSOD1, the majority of data are acquired using two chemical inhibitors, LCS1 and ATN-224, without supporting evidence that these inhibitors are acting in an on-target manner.

Overall, the authors address an important problem by seeking targetable vulnerabilities in PPM1D mutant AML cells, it is clear SOD1 deletion induces strong growth defects in the AML cell lines tested, most of the approaches are appropriate for the outcomes being evaluated, and the data are technically solid and well-presented. The major weakness lies in which redox pathways and ROS species are evaluated, how the resulting data are interpreted, and

gaps in the follow-up experiments. Due to these omissions, as currently presented, the broader impact of these findings are unclear.

These specific concerns are outlined in detail below and I offer some suggestions regarding how to clarify the mechanisms underlying their initial observation of SOD1 synthetic lethality:

(1) Fig. 1 - SOD1 appears to be clustered with several other genes in the volcano plot (including FANC proteins). Did any other ROS-detoxifying enzymes show similar fitness scores? The effects of the SOD1 sgRNA are striking, however it would be useful to see qPCR or immunoblot data confirming robust depletion.

Does SOD1 co-expression in PPM1-mutant patient AML correspond to poorer disease outcomes? This can be evaluated in publicly available patient datasets and would support the idea of SOD1 synthetic lethality.

It would also be useful to know (given the subsequent results) whether expression of the SOD2, the mitochondrial superoxide dismutase, is altered in response to SOD1 loss.

(2) Fig. 2 - What are the relative SOD1 levels in the mutant PPM1D vs. wt. cell lines? The effects of the chemical inhibitors are stronger in MOLM-13 than the other two lines. These data could also point to whether LCS-1 and ATN-224 cytotoxicity is on-target or off-target at these concentrations, which is a key issue not currently addressed in these studies. This is a particular concern as the OCI-AML2 line shows a stronger growth defect with CRISPR SOD1 KO (in Fig 1) but the smallest effects with these chemical inhibitors.

While endogenous mitochondrial superoxide levels are elevated in PPM1D mutant lines, it is entirely unclear why SOD1 inhibition should affect mitochondrial superoxide as it detoxifies cytosolic superoxide. Also unclear why DCFDA signal (which measures total hydroperoxides) is *increased* under SOD1 inhibition - SOD1 dismutates superoxide radicals into hydrogen peroxide, therefore unless SOD2 is compensating for SOD1 loss, one might expect hydroperoxides to be lower (unless some entirely different oxidase is increasing their levels). None of these outcomes appear to be considered. Finally, it is not explained how lipid peroxidation, which requires production of hydroxyl or similarly high potency radicals, is being caused by increased superoxide or peroxides. One possibility is there is an increase in labile iron, in which case this phenotype would be rescued by the iron chelator desferal, and by the lipophilic antioxidant, ferrostatin.

Do the sgSOD1 cells also show similar increases in MitoSox green, DCFDA and BODIPY signal? These experiments would clarify whether the effects with the inhibitors are directly related directly to SOD1 loss or if they represent off-target effects from the inhibitors and/or compensatory changes in SOD2.

(3) Fig. 3 - the effects on mitochondrial respiratory parameters, while statistically significant, do not seem biologically striking. Also, these data are shown for OCI-AML2 cells which show the smallest cytotoxic effects with the SOD1 inhibitors among the 3 lines tested. They do however show the most robust growth defect with sgSOD1. This discrepancy could suggest that mitochondrial dysfunction does not underlie the observed growth defect and/or the inhibitor cytotoxicity is not on-target. Ideally mitochondrial profiling should also be carried out on this cell line with inducible SOD1 depletion. Have the authors assessed whether the mitochondrial Bcl family proteins are affected by the inhibitors?

(4) Fig. 4 - Currently the data in this figure do not support the authors claim that PPM1D-mutant cells have impaired antioxidant defense mechanisms, leading to an elevation in ROS levels and reliance on SOD1 for protection. It should be noted that oxidative stress specifically refers to adverse cellular effects of increasing ROS, not baseline levels of various redox parameters. Ideally levels of GSSG/GSH would be a better measure of potential redox stress

tolerance than the total antioxidant capacity assay. Finally, oxidative stress can be assessed by challenging the wt and mutant PPM1D cell lines with oxidant stressors such as paraquat which elevates superoxide or drugs like erastin which elevate mitochondrial ROS. The immunoblot shows negligible changes in the antioxidant proteins assayed. Again, this blot should include SOD2 which is the most relevant antioxidant in the context of mitochondrial superoxide.

(5) Fig. 5 - These data support that DNA breaks are elevated in PPM1D mutant vs. wt cells. However, the data with the chemical SOD1 inhibitor again do not convince that the enhanced levels are due to on-target effects on SOD1. Use of the alkaline comet assay is appropriate for these studies and the 8-oxoguanine data do indicate contributions from oxidative DNA base damage. But these are unlikely to result directly from altered superoxide levels, as this species cannot directly oxidize DNA bases or cause DNA strand breaks.

The following points summarize my specific experimental and textual recommendations:

(1) These studies require an assessment of on-target efficacy of the inhibitors at the relevant concentration ranges. Ideally, they should have minimal effects against SOD1 knockout cell lines (acute challenge at a time point before the growth defects become apparent) and show better efficacy in SOD1-overexpressing lines. Key experiments (changes in superoxide, OCR profiling, DNA alkaline comet assay) would be more convincing if they are carried out with SOD1 knockout lines to compare against the inhibitor effects (3-4 days after introducing sgSOD1 when growth defects are not apparent).

(2) Instead of using NAC, which elevates glutathione synthesis but also has several known side-effects, the authors may want to determine whether Tempol, a SOD mimetic can rescue the effects of SOD1 knockout or inhibition. This would directly prove that SOD1 functional loss underlies the observed growth defect and cytotoxicity from genetic SOD1 knockdown or chemical inhibition.

(3) The complete lack of consideration of SOD2 in these studies is a missed opportunity as it reduces mitochondrial superoxide levels but elevates hydrogen peroxide levels. It would be very interesting to see whether SOD1 inhibition leads to compensatory increases in SOD2. SOD2 can be easily measured by immunoblot. Furthermore, measuring total superoxide via hydroethidium in a flow cytometric assay vs. mitochondrial ROS in PPM1D mut vs. wt cells and under SOD1 knockout would enable a determination of which species dominates (cytosolic or mitochondrial). These experiments are required to fill some logical gaps in interpretation of their redox data.

(4) Given the DNA breaks observed in PPM1D mutant cells, it is highly recommended the authors assess whether iron levels are elevated in mut vs. wt cells and whether desferal can rescue observed SOD1 inhibition defects.

(5) The authors may want to assess whether Rac1 or NADPH oxidase activity is altered in the SOD1 KO in wt vs. PPM1D cells. Their results may be the consequence of compromised ROS-driven survival signaling or DNA repair rather than direct ROS-induced damage, which is not caused directly by superoxide (or hydrogen peroxide).

(6) It is recommended the discussion focus more strongly on how the signaling function of superoxide vs. its reactions with other molecular entities to induce genotoxic outcomes could be contributing to the observed phenotypes. The discussion of FANC proteins, which were targets with similar fitness scores but not experimentally investigated at all, is an unwarranted digression.

<https://doi.org/10.7554/eLife.91611.2.sa1>

Reviewer #3 (Public Review):**Summary:**

Authors performed a genome-wide CRISPR-based screen for synthetic lethal interactions in leukemic cells expressing a mutant form of PPM1D and identified SOD1. Loss of SOD1 or its inhibition with small molecule compounds reduced survival of the cells containing truncated PPM1D. Further analysis revealed that mitochondria are functionally deficient in PPM1D mutant cells resulting in increased levels of ROS. Surprisingly, expression profiling and reverse phase protein arrays revealed that PPM1D mutant cells did not respond appropriately to the increased levels of ROS. The precise molecular mechanism underlying this phenotype remains currently unclear, nevertheless the study convincingly shows that PPM1D mutant cells are vulnerable to oxidative stress.

Strengths:

Experimental procedures used in the study are appropriate and overall the presented data are very convincing. The study identified an important vulnerability of leukemic cells that carry PPM1D mutation and provides a fundamental background for testing SOD1 inhibitors in preclinical research. In the revised version of the manuscript, authors provide several new experiments that support their former conclusions. In particular, they showed that deletion of SOD1 in AML cells improved survival of the transplanted mice and this effect was more prominent when using cells carrying the mutant PPM1D. Further, they included an important control experiment that showed decreased SOD1 activity after treatment with ATN-224 inhibitor.

Weaknesses:

In the opinion of reviewer, there are no obvious weaknesses in this study. In broader view, the findings presented here using in vitro cultures will need to be validated in vivo by future research. Cell lines used in the study were generated by CRISPR approaches in AML cells that have already been transformed. In addition, genome editing is inheritably connected with a risk of off target effects. It would therefore be great to identify AML samples carrying the PPM1D mutation that has been naturally selected during the transformation process.

<https://doi.org/10.7554/eLife.91611.2.sa0>

Author Response

The following is the authors' response to the original reviews.

We thank the reviewers for their insightful and constructive comments of our work that have helped to strengthen the manuscript. In response to the additional suggestions provided by the reviewers, we have made revisions by adding or replacing five main figures, three supplementary figures, refining the text, and clarifying certain conclusions. Detailed responses to the reviewers' points can be found below.

Additional experiments, textual changes, or modulation of claims are needed to address weaknesses in the SOD1 portion of the study. Specifically:

A) These studies require an assessment of the on-target efficacy of the inhibitors at the relevant concentration ranges. Ideally, they should have minimal effects against SOD1 knockout cell lines (an acute challenge at a time point before the growth defects become apparent) and show better efficacy in SOD1-overexpressing lines. Key experiments (changes in superoxide, OCR profiling, DNA alkaline comet assay) would be more

convincing if they were carried out with SOD1 knockout lines to compare against the inhibitor effects (3-4 days after introducing sgSOD1 when growth defects are not apparent). In addition, SOD activity should be measured directly following inhibitor treatment.

We agree with the reviewers that the on- vs. off-target effects of the pharmacologic SOD1 inhibitors is a critical point to address. We have validated that SOD activity is reduced following treatment with ATN-224 in Figure 2 – Figure supplement 1A.

Nevertheless, we acknowledge that the potential for off-target effects of these inhibitors cannot be completely ruled out. To address this concern, we have incorporated a discussion regarding the potential off-target effects of both LCS-1 and ATN-224.

B) Assays should be included to support that SOD1 activity is altered. ATN-224 and LCS-1 are used to inhibit SOD1 function in the majority of the experiments, which should be supported by SOD activity assays to confirm SOD inhibition. Further, the concentration of ATN-224 used in this paper (12.5 μ M) is beyond the concentration of what has been reported to inhibit SOD1 function in human blood cells. In Figure 4D, the authors demonstrate comparable SOD1 total protein levels in WT and PPM1D mutant cells. However, the authors should further address whether PPM1D-mutation alters SOD1 activity via SOD activity assays.

We thank the reviewers for these suggestions. We have performed SOD activity assays which confirmed that SOD activity is inhibited upon treatment with ATN-224 at two concentrations (6.25 and 12.5 μ M). Although we also did this for LCS-1-treated cells as well, in our hands, we did not see reduced SOD activity. However, LCS-1 has been shown to inhibit SOD activity in other publications including PMID: 21930909 and PMID: 32424294. From these assays, we have also found that PPM1D-mutant cells had increased SOD activity at baseline, despite having similar levels of SOD1 protein. These data have been added to Figure 2–Figure supplement 1A.

C) Some conclusions are not fully supported by the data provided. The authors claimed that "upon inhibition of SOD1, there was an increase in ROS that was specific to the mutant cells" in Figure 2E. Comparison of ROS levels among untreated, ATN-224, and LCS-1 of PPM1D-mutant cells should have been made and the statistics analysis among these groups should have been provided. Moreover, in Figure 2-Figure Supplement 1E, LCS-1 treatment does not increase ROS levels in PPM1D mutant LCLs. Performing these experiments with control and SOD1 deletion cells would have strengthened the results. Along with this point, the authors should comment on why SOD2 is not identified as a top hit in the CRISPR screen, as SOD2 deletion accumulates superoxide in cells.

After performing additional statistical analyses for Figure 2E, we found that the minor increase in ROS levels in the mutant cells after SOD1 inhibition was not statistically significant. We have revised the text accordingly.

As for why SOD2 was not identified as a top hit, we postulate that this may be due to inherent dependency of the WT cell lines on SOD2.

D) Fig. 1 - SOD1 appears to be clustered with several other genes in the volcano plot (including FANC proteins). Did any other ROS-detoxifying enzymes show similar fitness scores? The effects of the SOD1 sgRNA are striking, however, it would be useful to see qPCR or immunoblot data confirming robust depletion.

Thank you for your suggestion. We have validated the loss of SOD1 protein expression after SOD1 sgRNA deletion by immunoblot and have added this data to Figure 1–figure

supplement 1D. While other ROS-detoxifying enzymes were not significantly enriched in the top 37 hits, interestingly, the Fanconi Anemia pathway also has roles in counteracting oxidative stress. FA-deficient cells have mitochondrial dysfunction and redox imbalance, and several of the FA family proteins are implicated in mitophagy. Therefore, there may be an interesting interplay between SOD1 and the FA pathway that is worth highlighting in the discussion of our manuscript even though there was no experimental investigation performed.

E) Fig. 2 - What are the relative SOD1 levels in the mutant PPM1D vs. WT. cell lines? The effects of the chemical inhibitors are stronger in MOLM-13 than in the other two lines. These data could also point to whether LCS-1 and ATN-224 cytotoxicity are on-target or off-target at these concentrations, which is a key issue not currently addressed in these studies. This is a particular concern as the OCI-AML2 line shows a stronger growth defect with CRISPR SOD1 KO (in Fig 1) but the smallest effects with these chemical inhibitors. The authors should also include SOD1 levels for Figure 1D and Figure 4Figure supplement 1C.

SOD1 protein expression is similar between WT and PPM1D-mutant cell lines and the loss of SOD1 after SOD1 sgRNA deletion was validated by immunoblot. These data have been added to Figure 1- figure supplement 1D and Figure 4D.

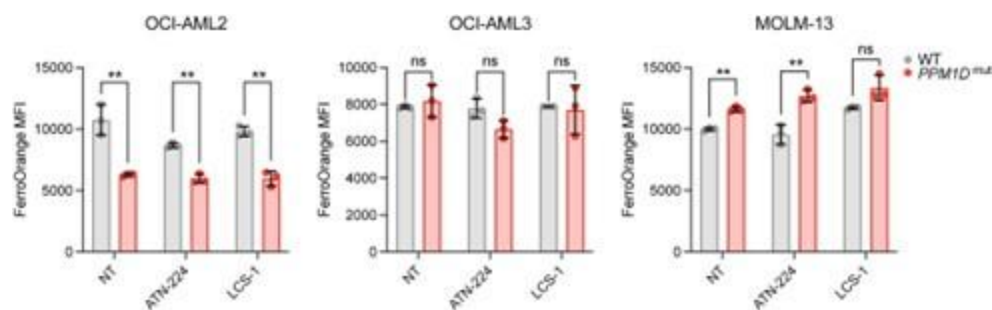
F) Does SOD1 co-expression in PPM1D-mutant patient AML correspond to poorer disease outcomes? This can be evaluated in publicly available patient datasets and would support the idea of SOD1 synthetic lethality.

Unfortunately, there are no publicly available patient datasets with sufficient cases of de novo PPM1D-mutant AML to assess this question.

G) While endogenous mitochondrial superoxide levels are elevated in PPM1D mutant lines, it is entirely unclear why SOD1 inhibition should affect mitochondrial superoxide as it detoxifies cytosolic superoxide. Also unclear why the DCFDA signal (which measures total hydroperoxides) is increased under SOD1 inhibition - SOD1 dismutates superoxide radicals into hydrogen peroxide, therefore unless SOD2 is compensating for SOD1 loss, one might expect hydroperoxides to be lower (unless some entirely different oxidase is increasing their levels). None of these outcomes appear to be considered. Finally, it is not explained how lipid peroxidation, which requires the production of hydroxyl or similarly high-potency radicals, is being caused by increased superoxide or peroxides. One possibility is there is an increase in labile iron, in which case this phenotype would be rescued by the iron chelator desferal, and by the lipophilic antioxidant, ferrostatin.

We measured intracellular labile iron levels by flow cytometry by staining the cells with FerroOrange at baseline and after SOD1 inhibition with our pharmacologic inhibitors (ATN-224 at 12.5 uM and LCS-1 at 1.25 uM). Across the three leukemia cell lines, we saw variable results in iron levels with no appreciable patterns (see below). Therefore, we cannot make conclusions about the contribution of labile iron to our observed phenotypes.

Author response image 1.



H) Do the sgSOD1 cells also show similar increases in MitoSox green, DCFDA, and BODIPY signal? These experiments would clarify whether the effects of the inhibitors are directly related directly to SOD1 loss or if they represent off-target effects from the inhibitors and/or compensatory changes in SOD2.

We do not observe changes in SOD2 in the several contexts in which we have examined this. We cannot exclude off-target effects of the inhibitors so have clarified this in the text.

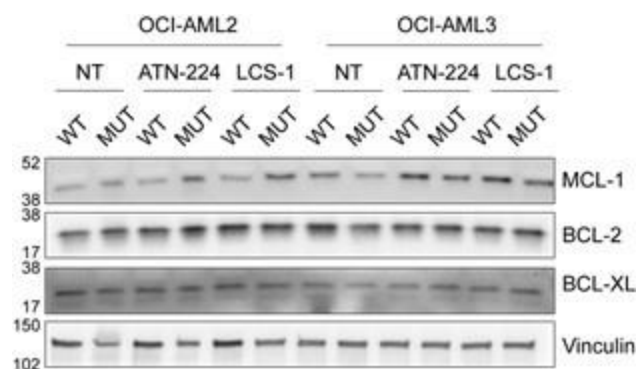
I) The authors may want to assess whether Rac1 or NADPH oxidase activity is altered in the SOD1 KO in WT vs. PPM1D cells. Their results may be the consequence of compromised ROS-driven survival signaling or DNA repair rather than direct ROS-induced damage, which is not caused directly by superoxide (or hydrogen peroxide).

We appreciate the reviewer's recommendations. However, due to time constraints, we regret not being able to assess Rac1 or NADPH oxidase activity. Nevertheless, we recognize the possibility of altered ROS-driven signaling rather than ROS-induced damage as a driver of our phenotype and have incorporated this possibility into our discussion.

J) Fig. 3 - the effects on mitochondrial respiratory parameters, while statistically significant, do not seem biologically striking. Also, these data are shown for OCI-AML2 cells which show the smallest cytotoxic effects with the SOD1 inhibitors among the 3 lines tested. They do however show the most robust growth defect with sgSOD1. This discrepancy could suggest that mitochondrial dysfunction does not underlie the observed growth defect and/or the inhibitor cytotoxicity is not on-target. Ideally, mitochondrial profiling should also be carried out on this cell line with inducible SOD1 depletion. Have the authors assessed whether the mitochondrial Bcl family proteins are affected by the inhibitors?

We assessed a few members of the mitochondrial Bcl-family proteins including MCL-1, BCL-2, and BCL-XL during the revision process. PPM1D-mutant cells have mildly increased expression of these anti-apoptotic proteins at baseline and the expression is not altered by pharmacologic SOD1 inhibition (see Author response image 2 below). Due to time constraints, we were unable to perform seahorse assays and mitochondrial profiling in the SOD1-deletion cells.

Author response image 2.



K) Fig. 4 - Currently the data in this figure do not support the authors' claim that PPM1D-mutant cells have impaired antioxidant defense mechanisms, leading to an elevation in ROS levels and reliance on SOD1 for protection. It should be noted that oxidative stress specifically refers to adverse cellular effects of increasing ROS, not baseline levels of various redox parameters. Ideally, levels of GSSG/GSH would be a better measure of potential redox stress tolerance than the total antioxidant capacity assay. Finally, oxidative stress can be assessed by challenging the wt and mutant PPM1D cell lines with oxidant stressors such as paraquat which elevates superoxide, or drugs like erastin which elevate mitochondrial ROS. The immunoblot shows negligible changes in the antioxidant proteins assayed. Again, this blot should include SOD2 which is the most relevant antioxidant in the context of mitochondrial superoxide.

We measured intracellular glutathione levels by flow cytometry and found that PPM1D-mutant cells had a greater proportion of cells with low levels of GSH. This data has been added as Figure 4D. We have also repeated the western blot to look at the antioxidant proteins catalase, SOD1, and thioredoxin after SOD1-deletion and pharmacologic SOD1 inhibition. We evaluated SOD2 protein levels in these experiments, as suggested. Smooth muscle actin (SMA) is included in the antibody cocktail as a loading control. However, it is unclear to us as to why PPM1D-mutant cells consistently have significantly higher levels of SMA. Therefore, we included a separate loading control, Vinculin. Repeat of these western blots showed a clearer difference between WT and PPM1D-mutant cells in the levels of these antioxidant proteins in which PPM1D-mutant cells have decreased levels of catalase and thioredoxin. These blots also show that SOD2 levels may be mildly increased in the PPM1D-mutant cells at baseline but is not significantly upregulated upon SOD1 inhibition. We have replaced the original immunoblot from Figure 4D with the revised blots that more clearly demonstrate the reduced levels of catalase and thioredoxin, now figure 4E.

L) Fig. 5 - These data support that DNA breaks are elevated in PPM1D mutant vs. wt cells. However, the data with the chemical SOD1 inhibitor again do not convince us that the enhanced levels are due to on-target effects on SOD1. Use of the alkaline comet assay is appropriate for these studies and the 8-oxoguanine data do indicate contributions from oxidative DNA base damage. But these are unlikely to result directly from altered superoxide levels, as this species cannot directly oxidize DNA bases or cause DNA strand breaks.

Thank you to the reviewers for raising this point. We have performed comet assays in SOD1-deletion cells to look at levels of DNA damage. Consistent with the reviewers' point, we do not see a significant increase in DNA breaks after SOD1 deletion. We have removed the data

using the SOD1 inhibitor and instead show the COMET analysis in the PPM1D-mut and SOD1-KO cells (see Figure 5F). We now make the point that increased DNA damage with SOD1 loss cannot explain the vulnerability of the double-mutant cells.

M) Instead of using NAC, which elevates glutathione synthesis but also has several known side effects, the authors may want to determine whether Tempol, a SOD mimetic can rescue the effects of SOD1 knockout or inhibition. This would directly prove that SOD1 functional loss underlies the observed growth defect and cytotoxicity from genetic SOD1 knockdown or chemical inhibition.

This is an excellent suggestion; we have added comments to this effect into the discussion.

N) It is recommended the discussion focus more strongly on how the signaling function of superoxide vs. its reactions with other molecular entities to induce genotoxic outcomes could be contributing to the observed phenotypes. The discussion of FANC proteins, which were targets with similar fitness scores but not experimentally investigated at all, is an unwarranted digression.

Thank you for this recommendation. We have expanded the discussion to focus more on the signaling functions of superoxide. However, considering the role of the Fanconi Anemia pathway in mitigating DNA damage and oxidative stress, we believe the discussion on the FANC proteins is important due to the possible intersection with SOD1. Therefore, we have refined this portion discussion to focus more on the interplay between SOD1 and FA.

O) The complete lack of consideration of SOD2 in these studies is a missed opportunity as it reduces mitochondrial superoxide levels but elevates hydrogen peroxide levels. It would be very interesting to see whether SOD1 inhibition leads to compensatory increases in SOD2. SOD2 can be easily measured by immunoblot. Furthermore, measuring total superoxide via hydroethidium in a flow cytometric assay vs. mitochondrial ROS in PPM1D mut vs. wt cells and under SOD1 knockout would enable a determination of which species dominates (cytosolic or mitochondrial). These experiments are required to fill some logical gaps in the interpretation of their redox data.

During the revision process, we have included SOD2 in our studies and have found that loss of SOD1 via genetic deletion and pharmacologic inhibition does not lead to compensatory increases in SOD2 (Figure 4D). Additionally, we have measured cytoplasmic superoxide levels using dihydroethidium to differentiate between cytoplasmic vs. mitochondrial superoxide. We found that at baseline levels, the mutant cells also harbored more cytoplasmic superoxide. We have added this figure as Figure 2C and moved the original mitochondrial superoxide data to Figure 2-figure supplement 1C.

P) Given the DNA breaks observed in PPM1D mutant cells, it is highly recommended that the authors assess whether iron levels are elevated in mut vs. wt cells and whether desferal can rescue observed SOD1 inhibition defects. Also, it has been reported that PPM1D promotes homologous recombination by forming a stable complex with BRCA1-BARD1, thereby enhancing their recruitment to doublestrand break sites. The authors should comment on why there is no difference in repair via HR in WT and PPM1D mutant cells in Figure 5C.

Please see comment G regarding our findings about iron levels.

The reviewers pose an interesting question as to why there is no difference in HR repair between WT and mutant cells, given the reported role of PPM1D in promoting HR. We have

addressed this question in the main text. We believe that several factors can limit the extent of HR enhancement in PPM1D-mutant cells. For example, HR is typically confined to the S/G2 phase and thus may be constrained by cell cycling, among other regulatory mechanisms.

Other comments:

A) The authors described in the Method section that "The CRISPR Screen PPM1D mutant Cas9expressing OCI-AML2 cell lines were transduced with lentivirus library supernatant." The authors need to provide information on whether the MOI of the CRISPR screen has been well controlled to ensure that the majority of the cell population has a single copy of sgRNA transduction.

We performed a lentiviral titer curve prior to the screen to determine the volume of viral supernatant to add for a multiplicity of infection (MOI) of 0.3. This important detail has been added to our Methods.

B) The study convincingly shows differences between parental leukemic cells and the PPM1D mutants but one important control is missing in experiments related to Fig. 2 and 3. All PPM1D mutant clones used in this study were subjected to the blasticidin selection of the transduced cells to generate cells stably expressing Cas9 and subsequently, the clones with successful PPM1D targeting were expanded. The authors should demonstrate that increased ROS production is not just a consequence of the lentiviral transduction and antibiotic selection and that it corresponds to increased PPM1D activity in PPM1D mutant cells. To do that, authors could compare PPM1D clones to parental cells that underwent the same selection procedure (OCI-AML2-Cas9 cells and OCI-AML3-Cas9 cells).

It is true that the parental OCI-AML2 and OCI-AML3 cell lines underwent four days of blasticidin selection to create the stably expressing Cas9 cell lines. However, after the four-day period, the blasticidin was removed from the cell culture media. From there, we induced the PPM1D-mutations into the Cas9-expressing "WT" cell lines using the RNP-based CRISPR/Cas9 delivery method and single cells were then sorted into 96-well plates. Clones were expanded and validated using Sanger sequencing, TIDE analysis, and western blot. In all of our assays, we compare the WT Cas9 cells to the PPM1D-mutant Cas9 cells. Additionally, the cells have been expanded and passaged several times after blasticidin-selection. Therefore, we believe it is unlikely that there are residual ROSinducing effects from the antibiotic treatment.

C) The authors mention that they identified 3530 genes differentially expressed in parental and PPM1D mutant cells (line 267) but it is unclear what was the threshold for statistical significance. They mention FDR<0.05 in the Methods but show GSEA analysis with FDR<0.25 in Figure 4A. Source data for Fig. 4 is missing and the list of differentially expressed genes is not shown.

The source data files for Figures 1 and 4 will be uploaded with the revised manuscript. Upon reviewing the source data, we noticed an error in the number of differentially expressed genes. We have corrected this in line 274 and you will see that this correlates with Figure 4-source data 1. For the thresholds, we used an FDR<0.05 for the differential gene expression analysis, and an FDR <0.25 in the GSEA, which is an appropriate threshold for GSEA. We have clarified these thresholds in the methods section.

D) Include a definition of MFI in Figure legend Fig.2 and also in the Methods section. The unit should be indicated at both the x and y axes.

We have defined MFI in the figure legends and methods sections and have updated the figures accordingly.

E) Legend to Figure 2 - Figure Supplement 1 E should define the grey and pink columns (likely WT and mutants LCLs).

Thank you. We have defined the grey and pink columns as WT and PPM1D-mutant cell lines, respectively for Figure 2 – Figure supplement 2D and E.

F) Reporter assays in Fig. 5 convincingly show that NHEJ capacity is reduced in PPM1D mut cells. In the text, the authors state that this might reflect the impact of PPM1D on LSD1 (line 365). Although this might be the case, other options are equally possible. It would be appropriate to include a reference to the ability of PPM1D to counteract γH2AX and ATM which generate the most upstream signals in DDR.

Thank you to the reviewers for raising this excellent point. We have revised the text to incorporate the impact of PPM1D on γH2AX and ATM on NHEJ.

G) The authors correctly state that truncation of PPM1D leads to protein stabilization (line 85) and that it is present in U2OS cells (line 355). These observations have first been reported by Kleiblova et al 2013 and therefore one reviewer believes that this reference should be included. This study also identified truncating PPM1D mutation in colon adenocarcinoma. HCT116 cells and the role of PPM1D mutation in promoting the growth of colon cancer has subsequently been tested in an animal model (Burocziova et al., 2019).

Thank you. We have added this reference to our text in line 360.